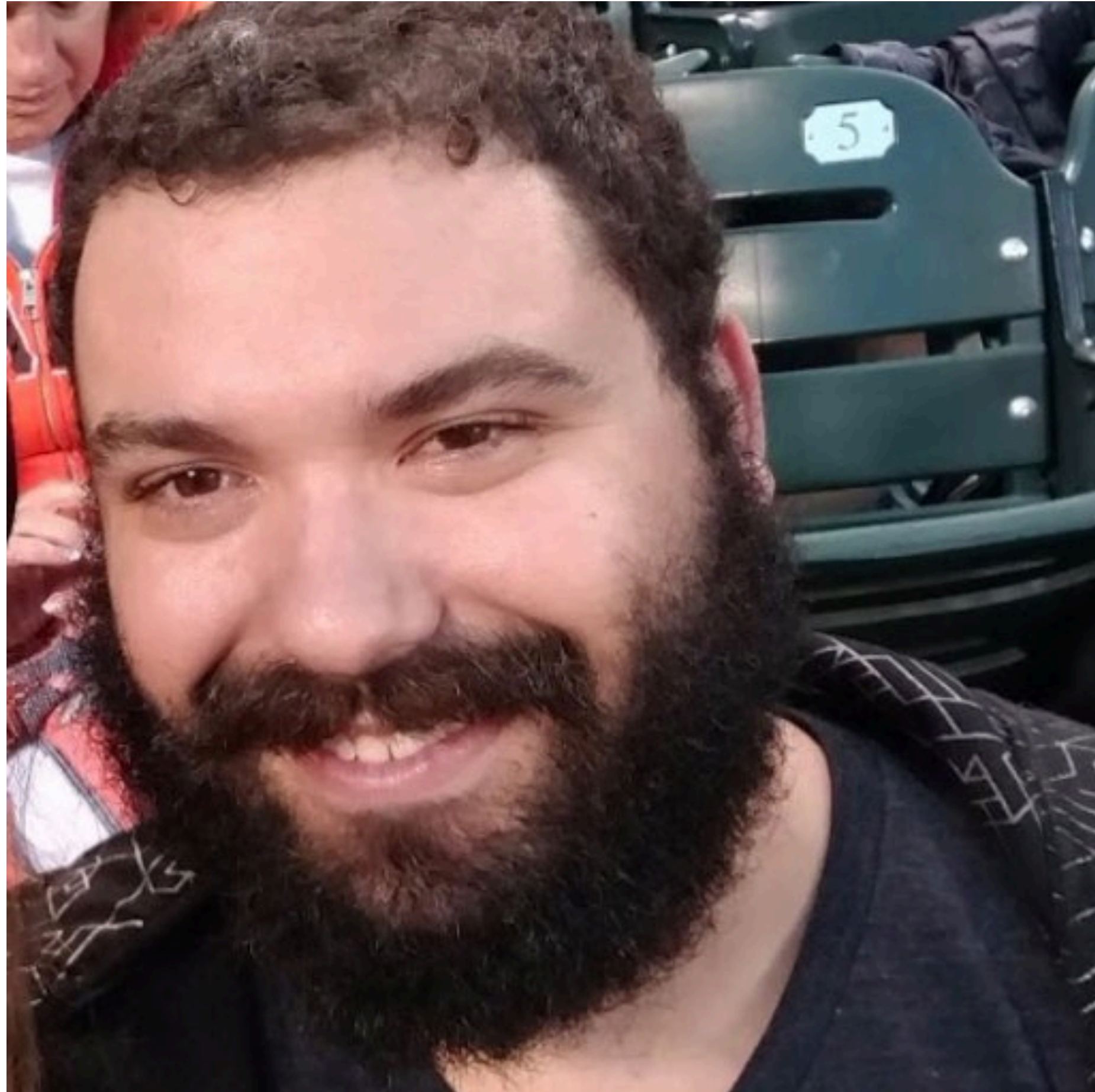


Workload Identity

Evan Gilman



About Me



Academia (Network Engineering)

Living in Silicon Valley

Author of "Zero Trust Networks"

Working on Workload Identity

Agenda

Zero Trust Basics

Identity in the Datacenter

SPIFFE

SPIRE Overview

SPIRE Walkthrough

Zero Trust Basics



ATTACKER CONTROLLED

Zero Trust Basics



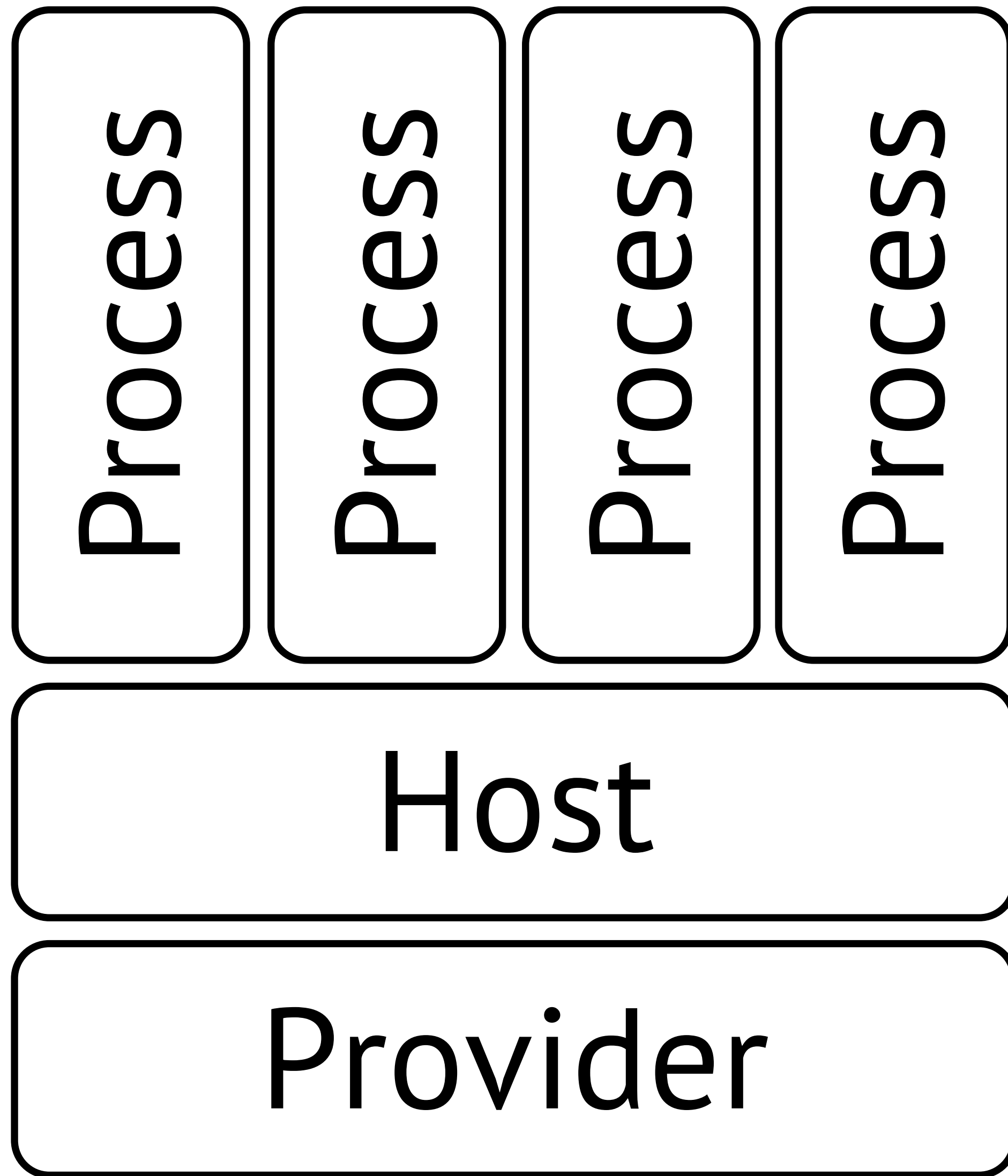
ATTACKER CONTROLLED
How to Build Reliable System?

Zero Trust Basics



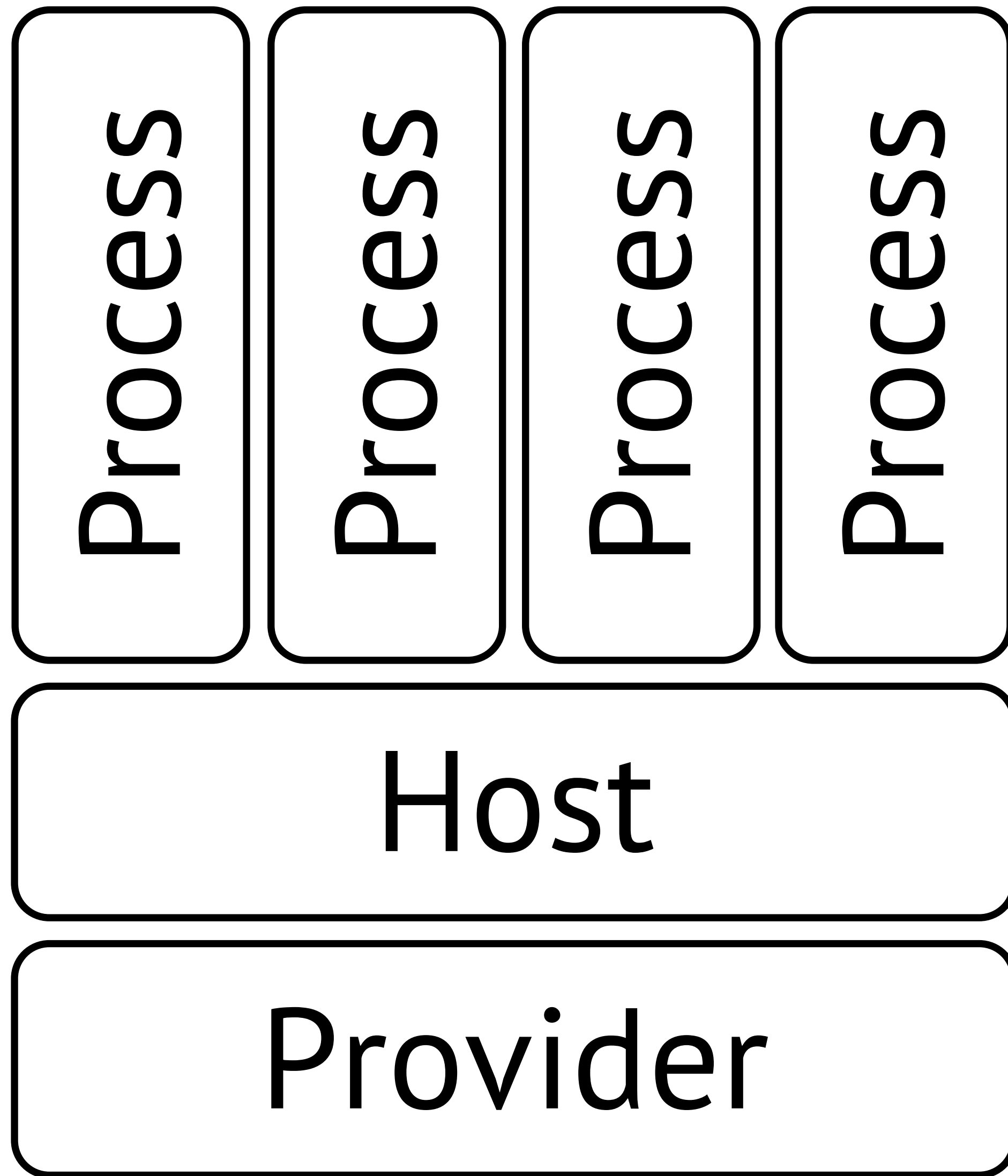
ATTACKER CONTROLLED

Authentication! But... Need Identity



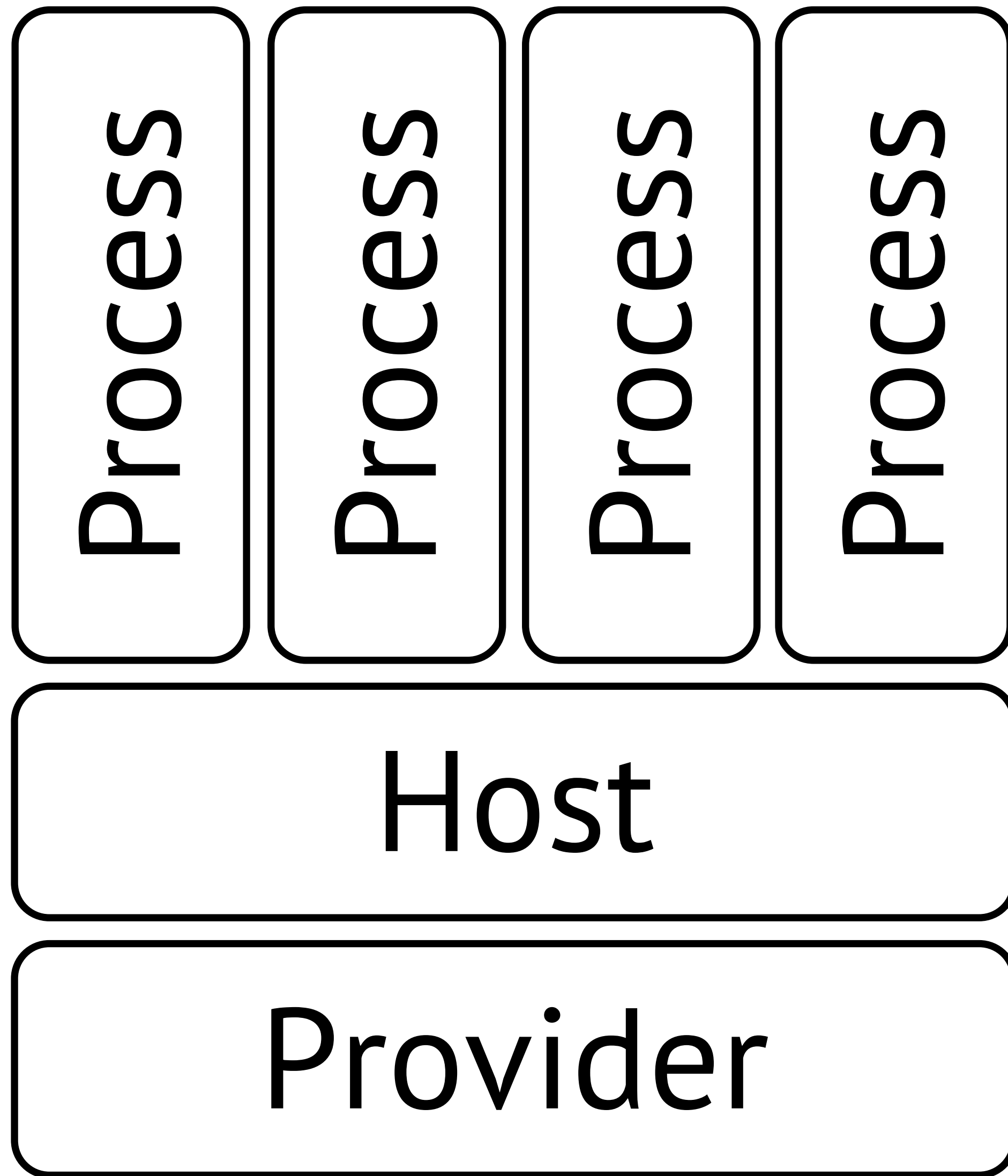
Security Group: sg-edcd9784

Vendor-Specific



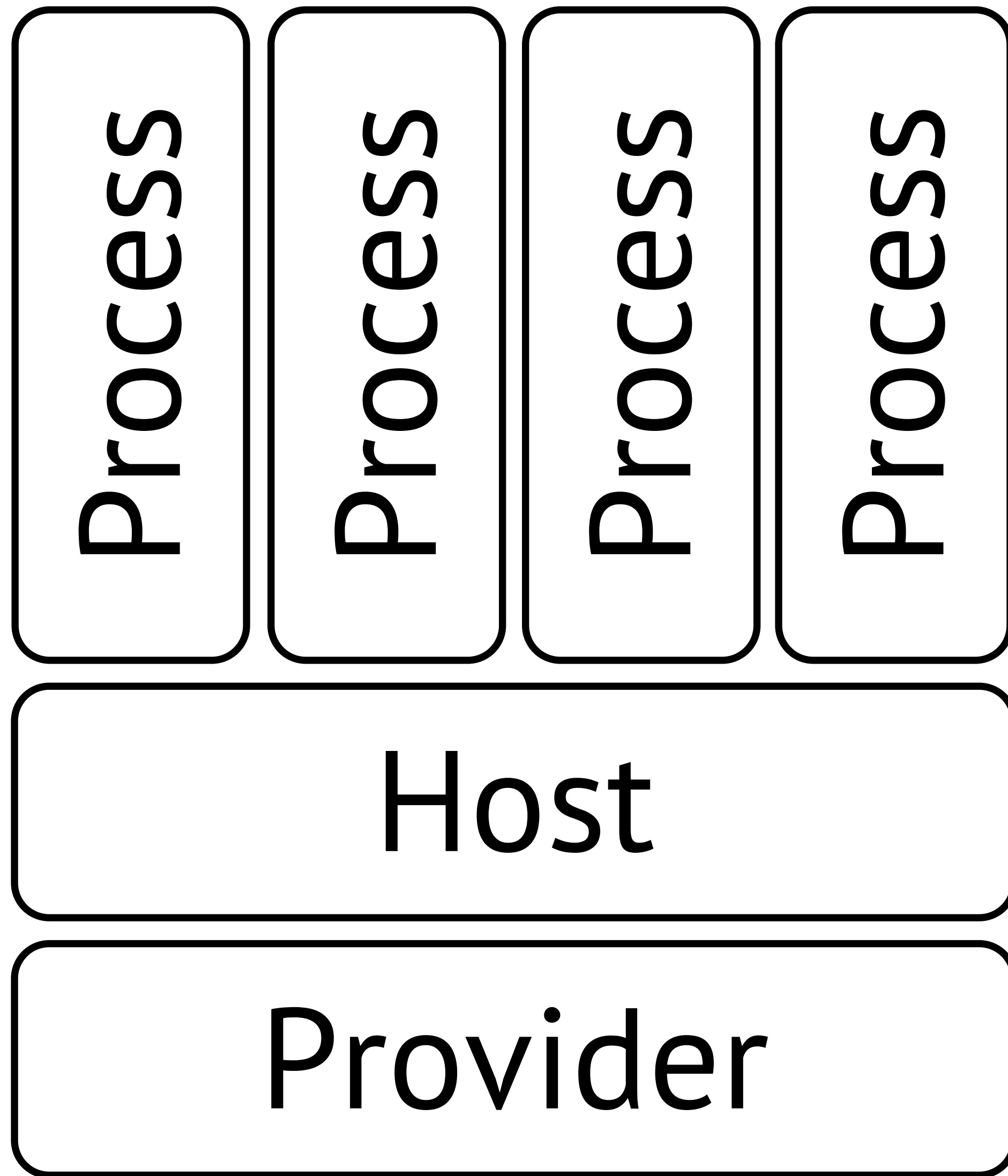
IP(s): 192.168.0.1/24

~~Security Group: sg-edcd9784~~



IP(s): 192.168.0.1/24

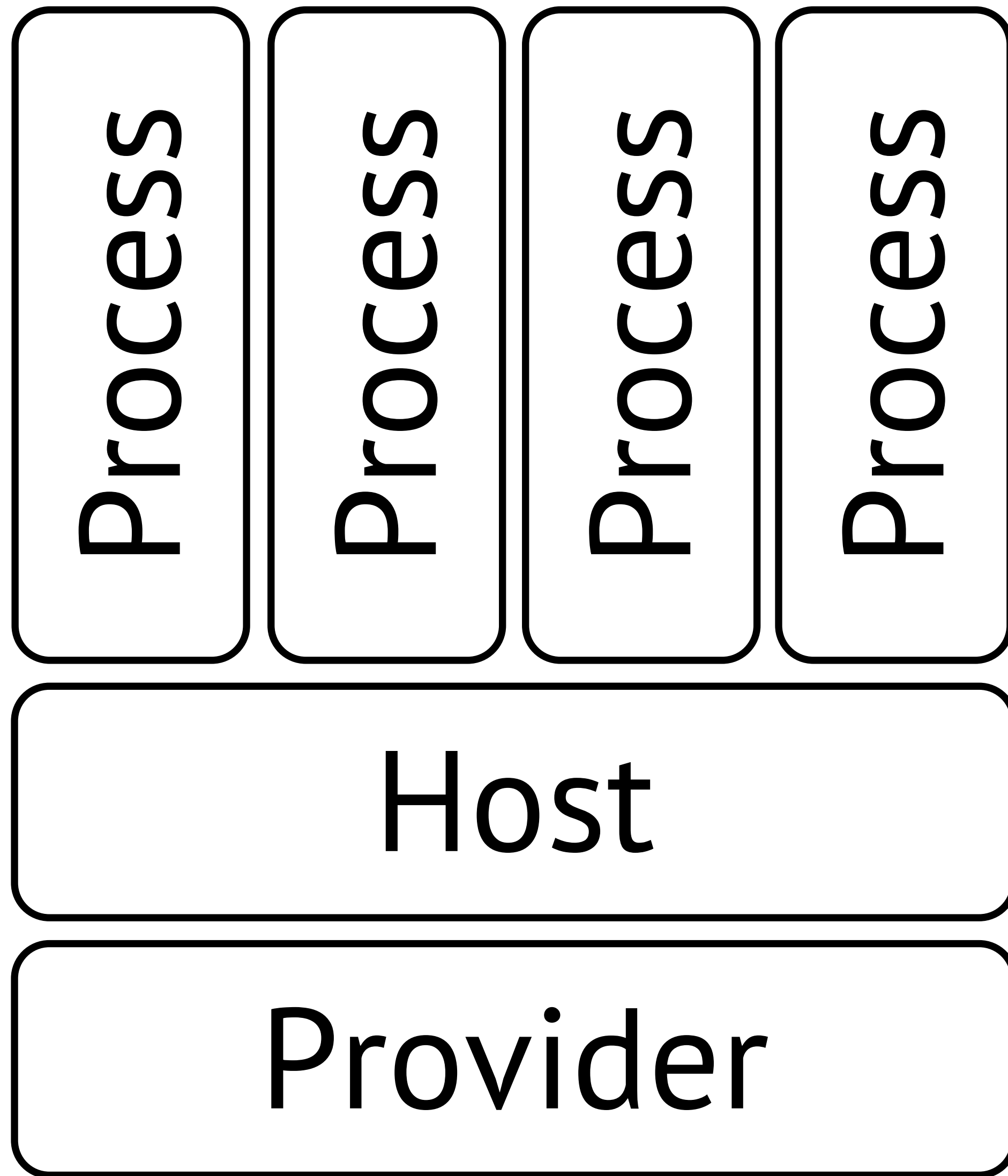
~~Security Group: sg-edcd9784~~



IP(s): 192.168.0.1/24

~~Security Group: sg-edcd9784~~

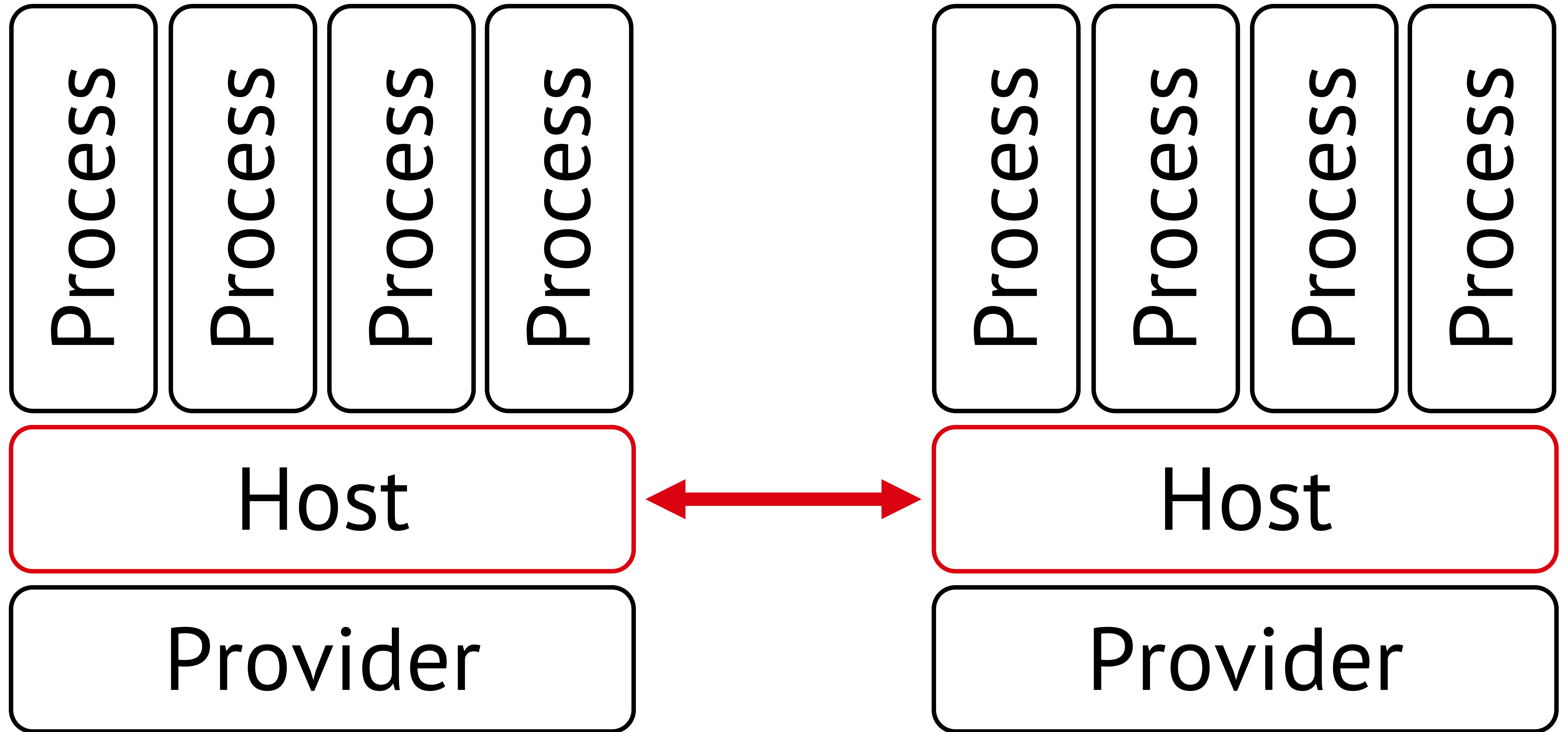
Can We Trust IP???

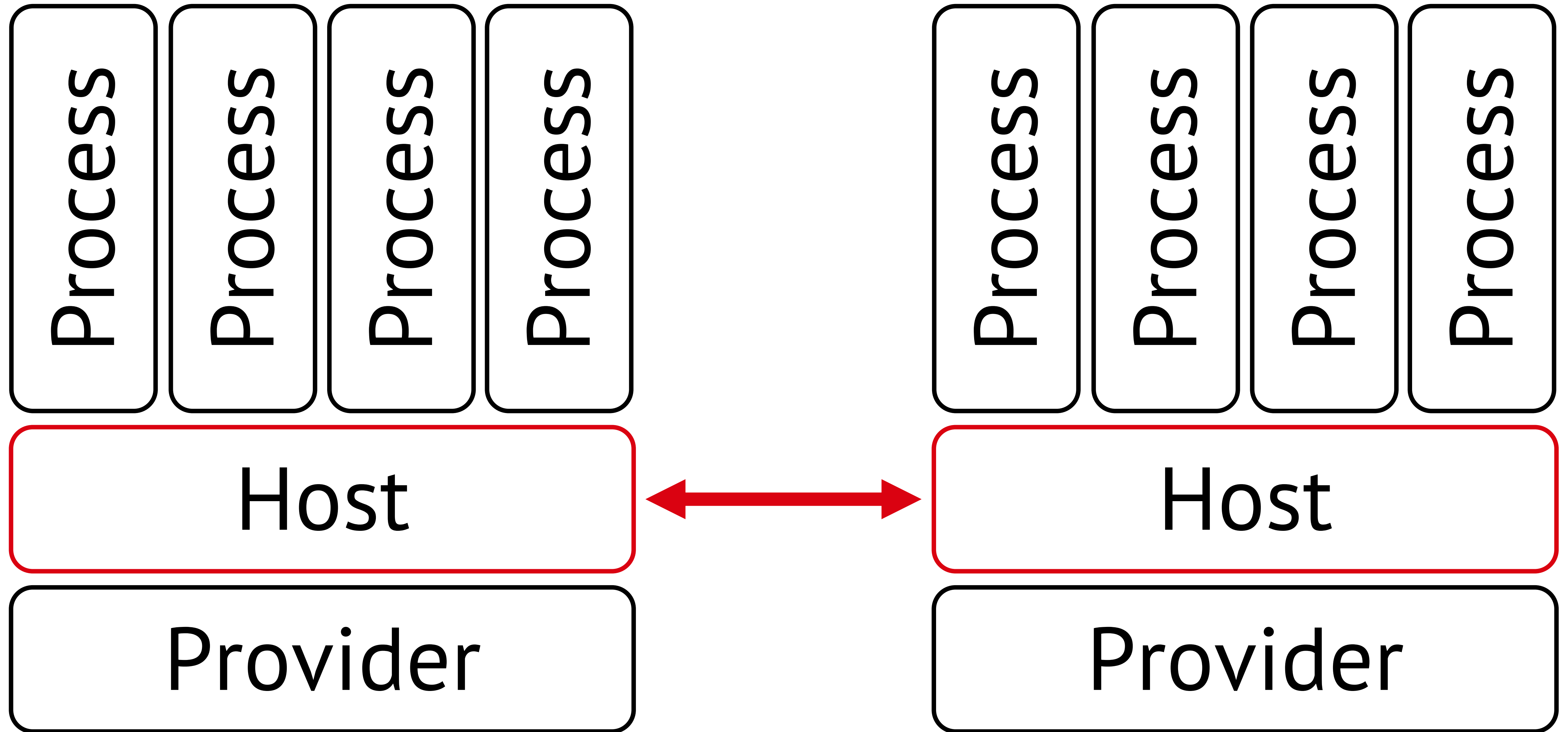


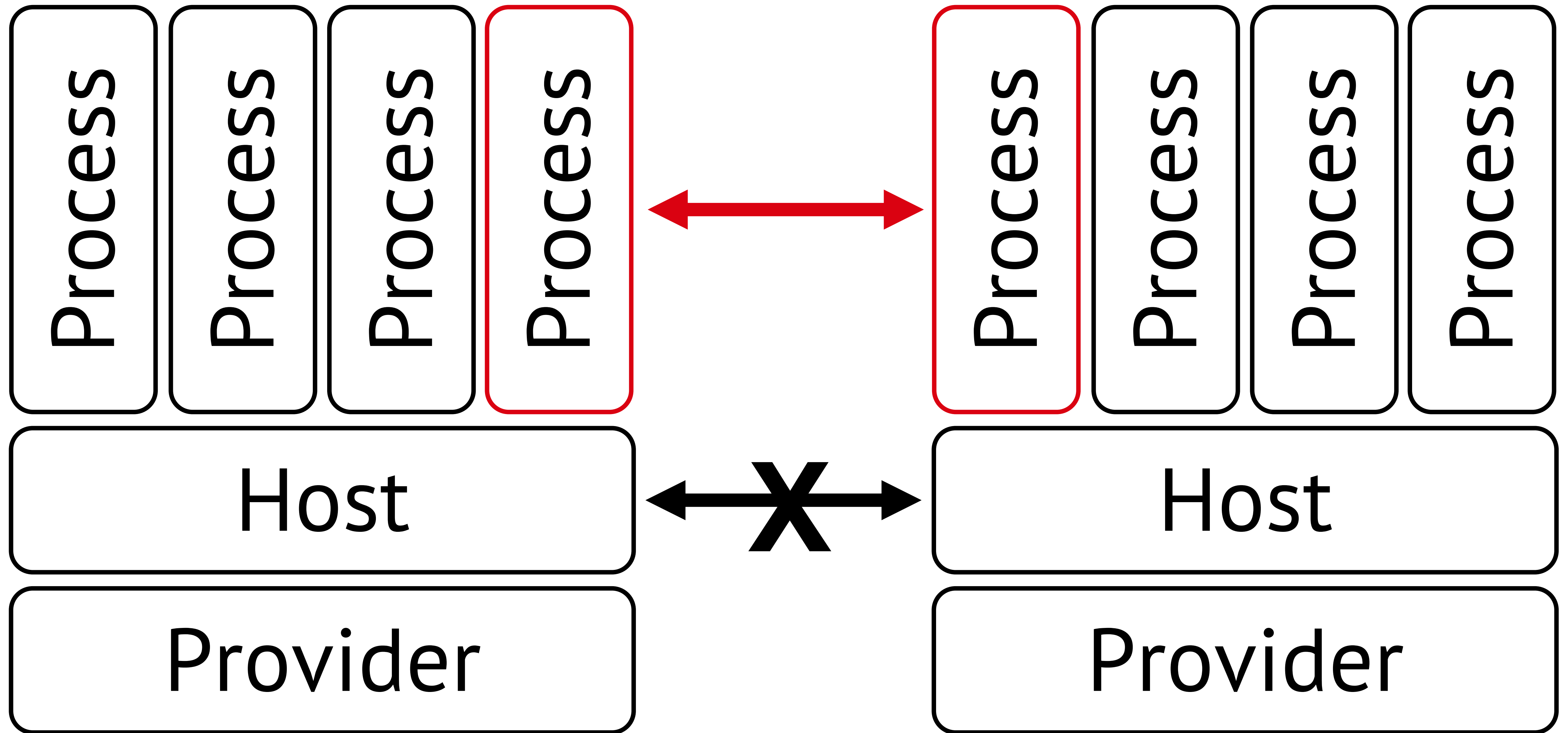
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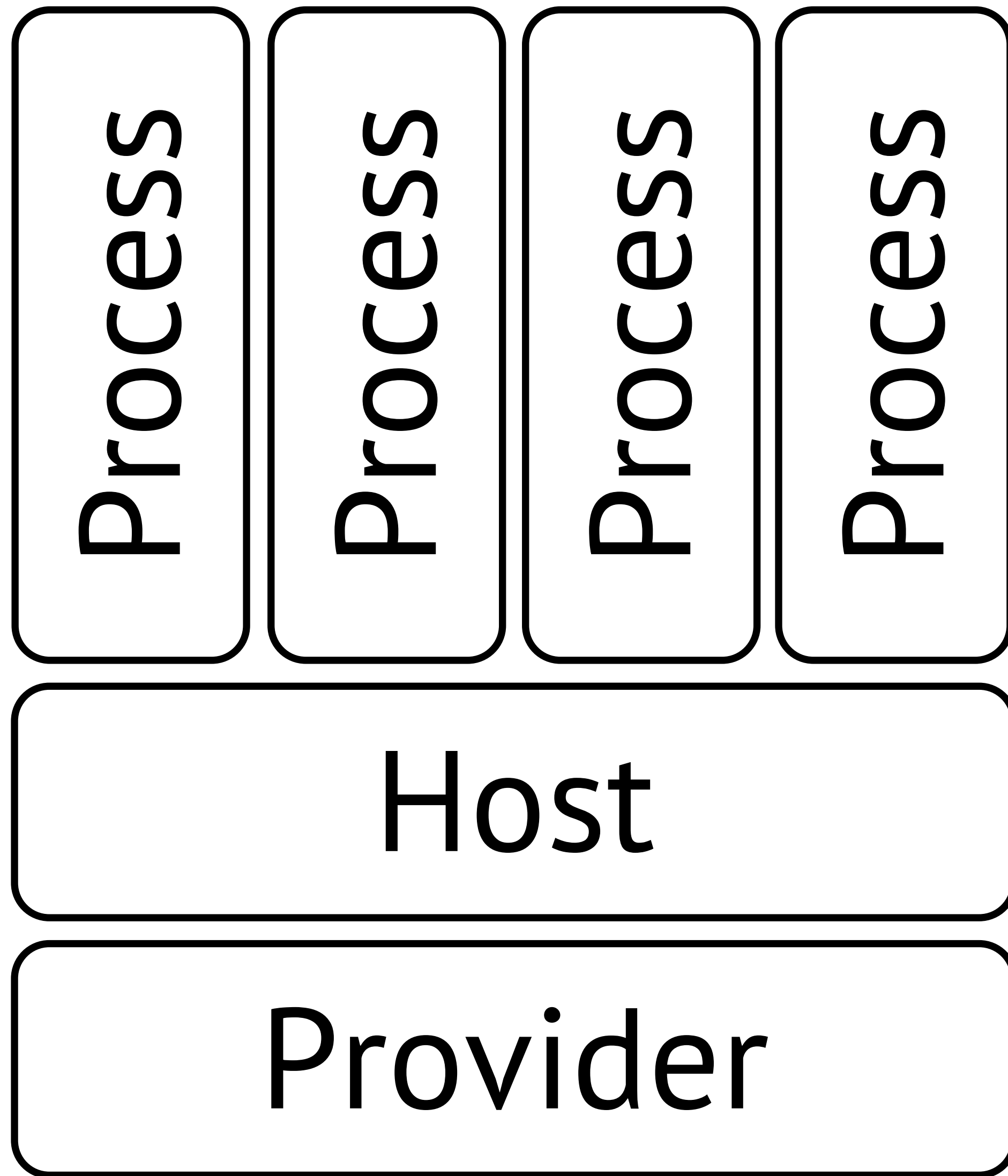
~~Security Group: sg-edcd9784~~

What Are We Trying to Model?



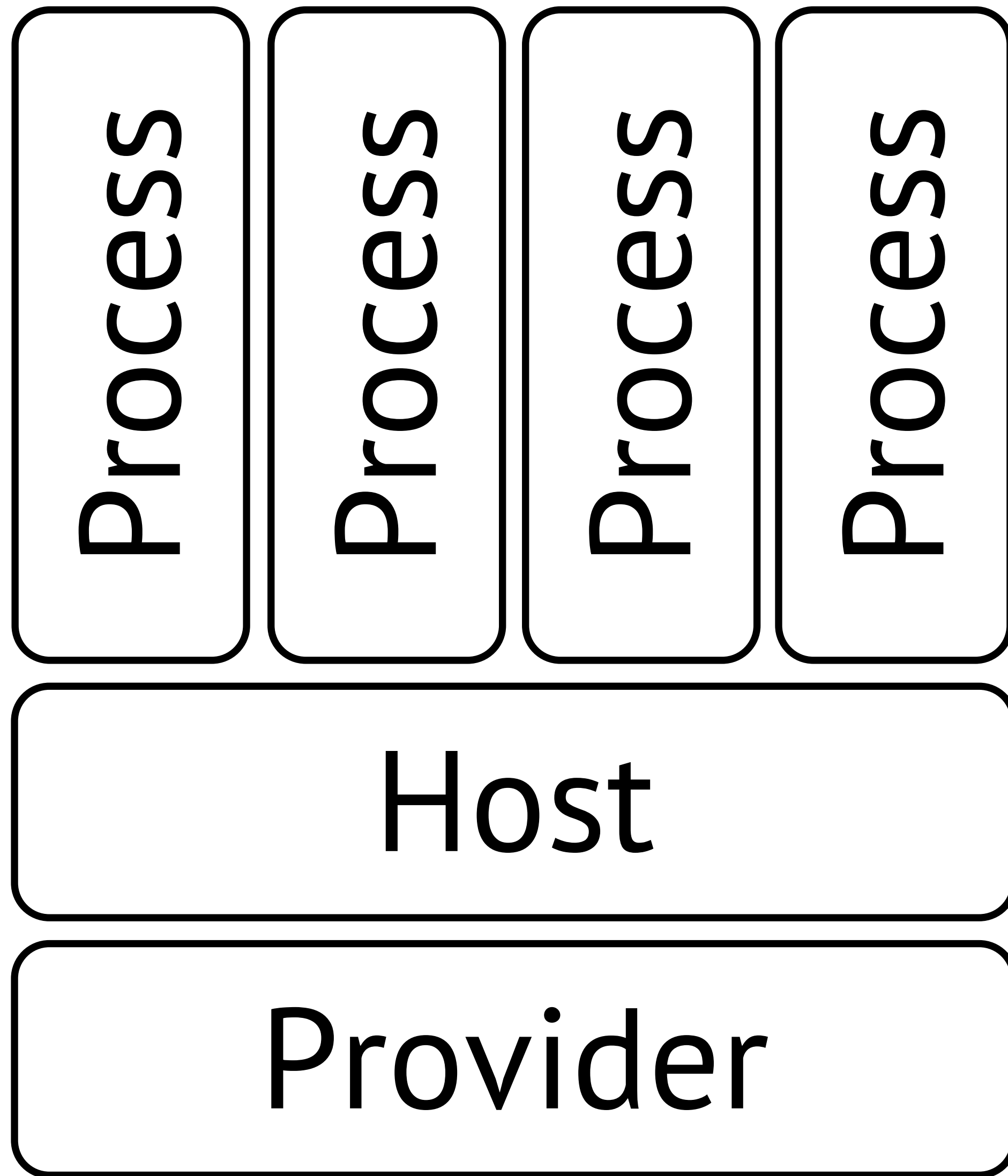






~~IP(s): 192.168.0.1/24~~

~~Security Group: sg-edcd9784~~



???

~~IP(s): 192.168.0.1/24~~

~~Security Group: sg-edcd9784~~

Workload Identity

Identity Domains

No Interoperability

Mesos

Hyper-V



Dell

HP



Kubernetes

KVM



Google Cloud Platform

Identity
Domain

Identity Domains

No Interoperability

Mesos

Hyper-V



Dell

HP



Kubernetes

KVM



Google Cloud Platform

Identity
Domain

Identity Domains

How to Bridge Domains?

Mesos

Hyper-V



Dell

HP



Kubernetes

KVM



Google Cloud Platform

Identity
Domain

Universal Workload Identity

SPIFFE

SPIFFE

**Identity Issuance, Validation,
– – – – Interoperability – – – –**

Single Standardized Identity Domain
Well-defined Authentication Semantics
Identity Issuance as Infrastructure Service

SPIFFE

**Identity Issuance, Validation,
– – – – Interoperability – – – –**

Single Standardized Identity Domain
Well-defined Authentication Semantics
Identity Issuance as Infrastructure Service

SPIFFE ID

spiffe://example.org/fo
o

SPIFFE ID

spiffe://example.org/fo

o

Trust Domain

SPIFFE ID

spiffe://example.org/fo

o

Workload Identifier

SPIFFE Verifiable Identity Document

spiffe://example.org/fo

o



SPIFFE Verifiable Identity Document

spiffe://example.org/fo

o



SPIFFE Verifiable Identity Document

spiffe://example.org/fo

o



SPIFFE Verifiable Identity Document

spiffe://example.org/fo

o



SPIFFE Verifiable Identity Document

spiffe://example.org/fo

o



SPIFFE Verifiable Identity Document

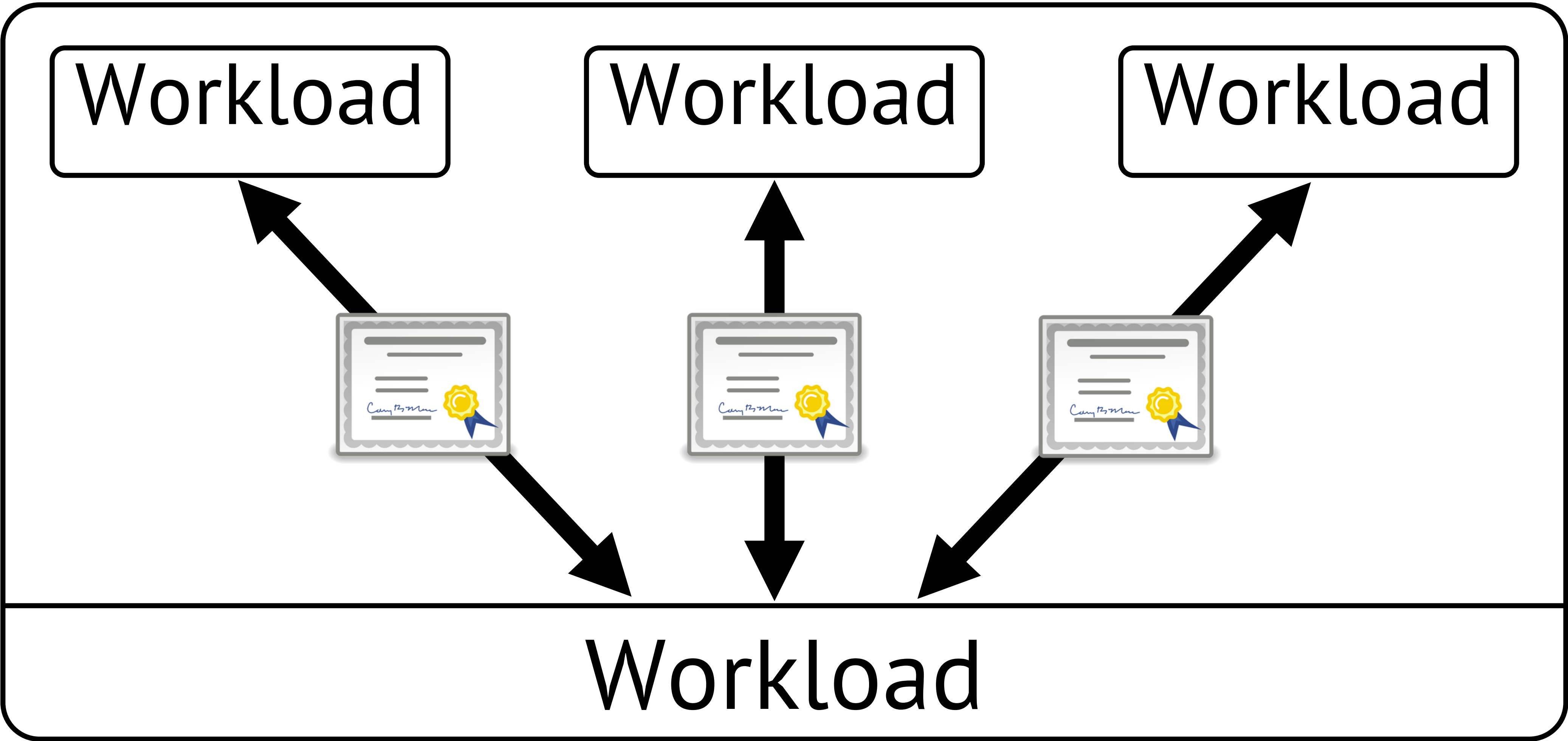
spiffe://example.org/fo

o



SPIFFE Workload API

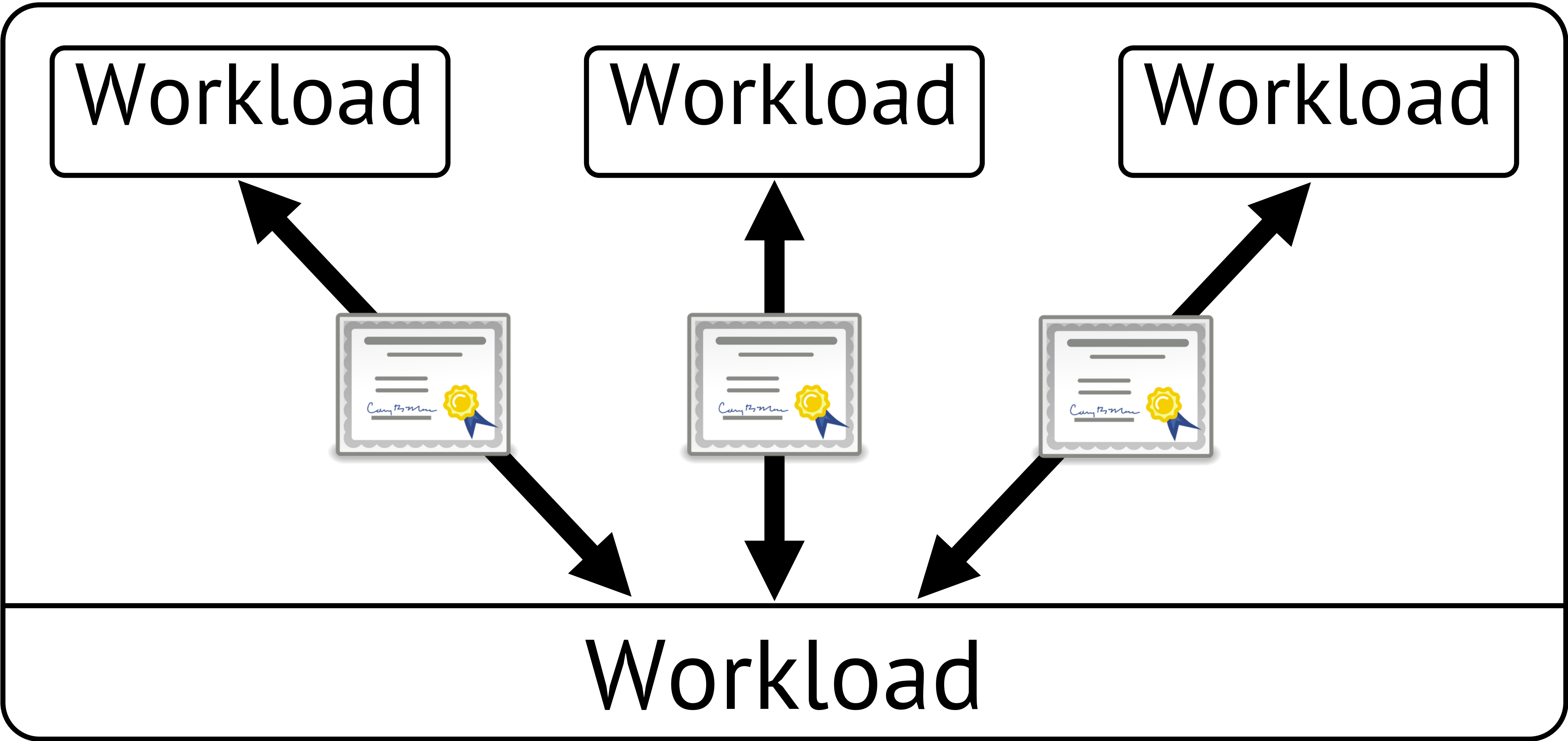
Server



Local API Provider Provides Identity

SPIFFE Workload API

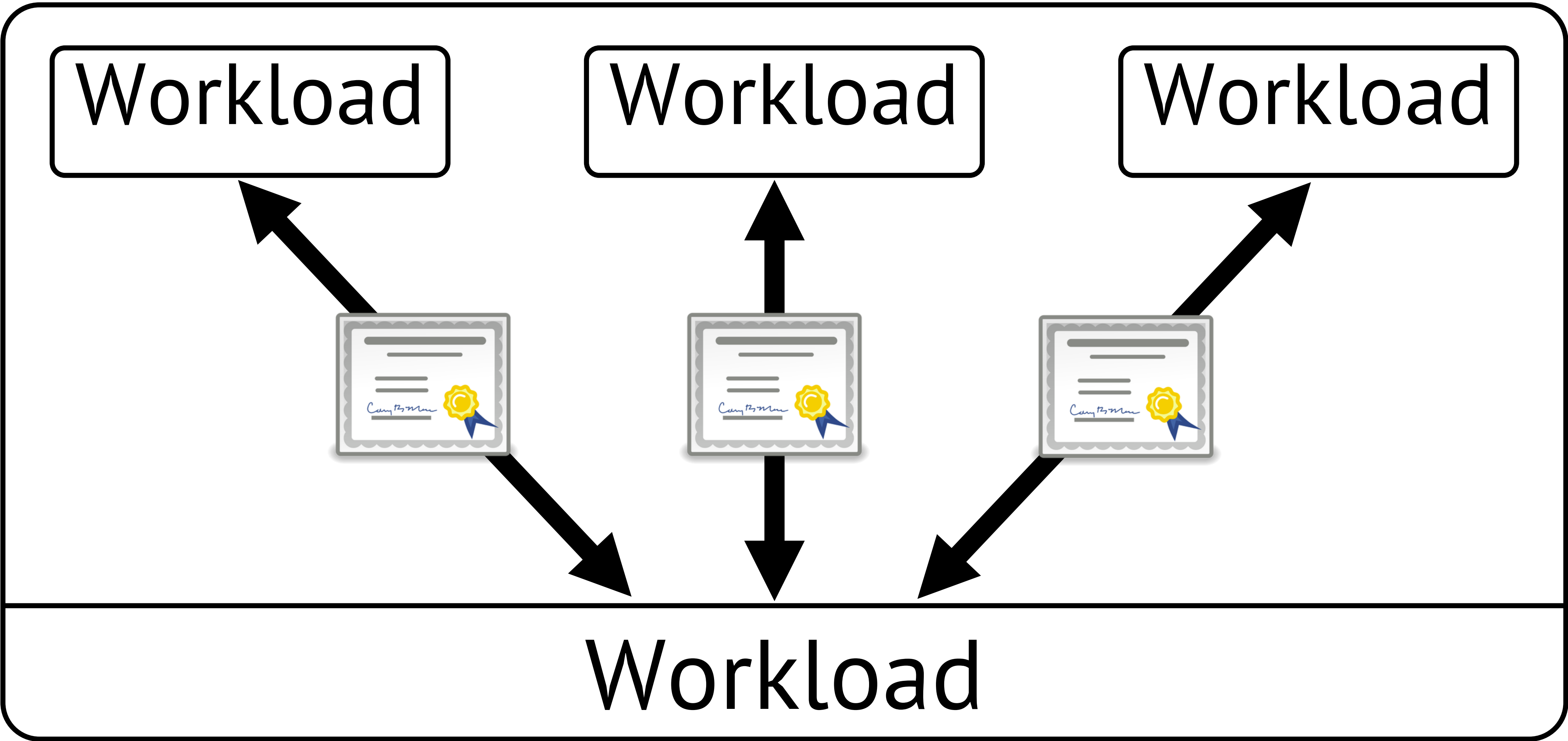
Server



Identity Available on any Platform

SPIFFE Workload API

Server



Identity Available on any Platform

How Do I SPIFFE?

SPIRE

SPIRE

SPIRE

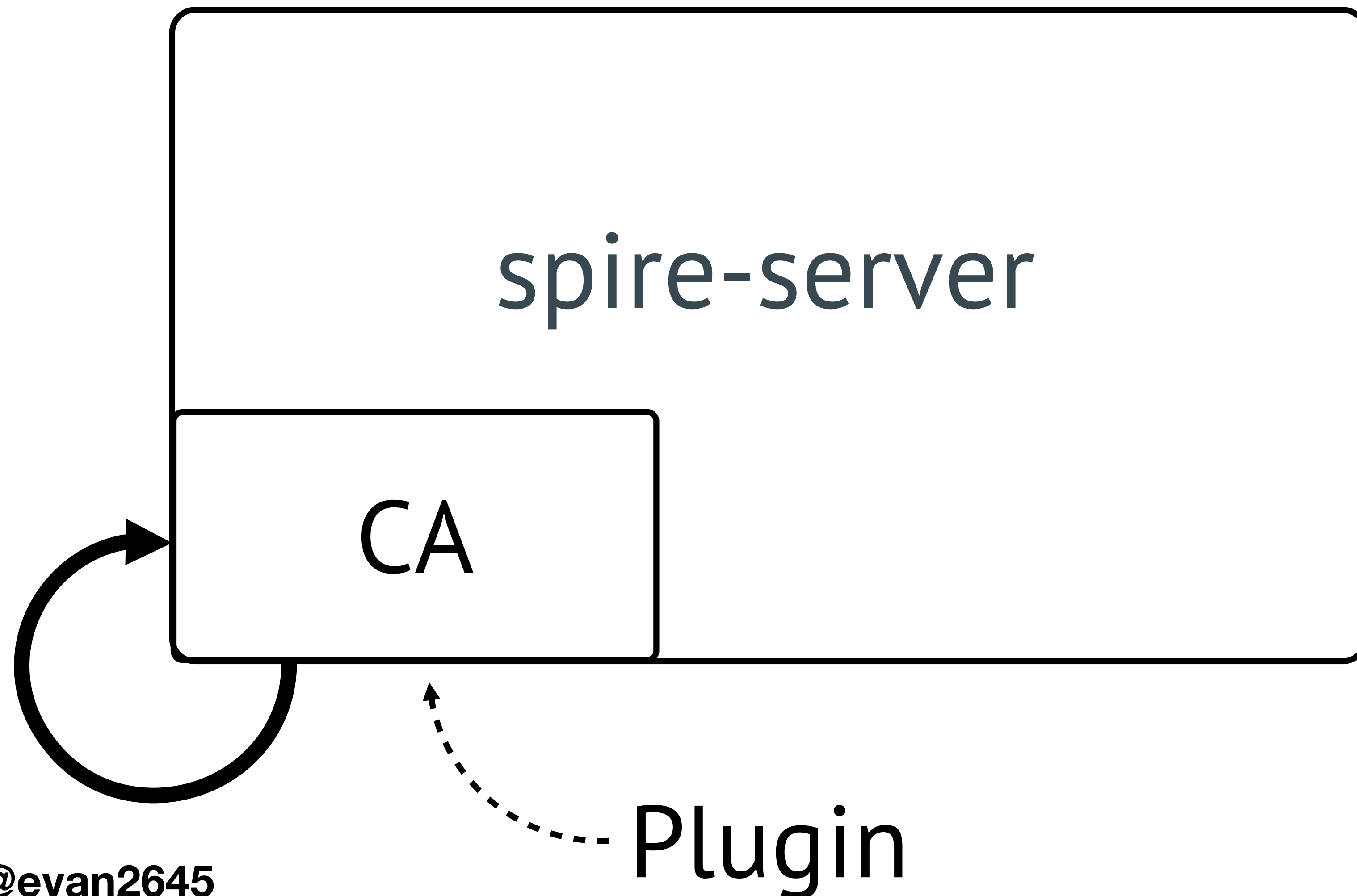
spire-server

- Identity Mapping
- Node Attestation
- SVID Issuance

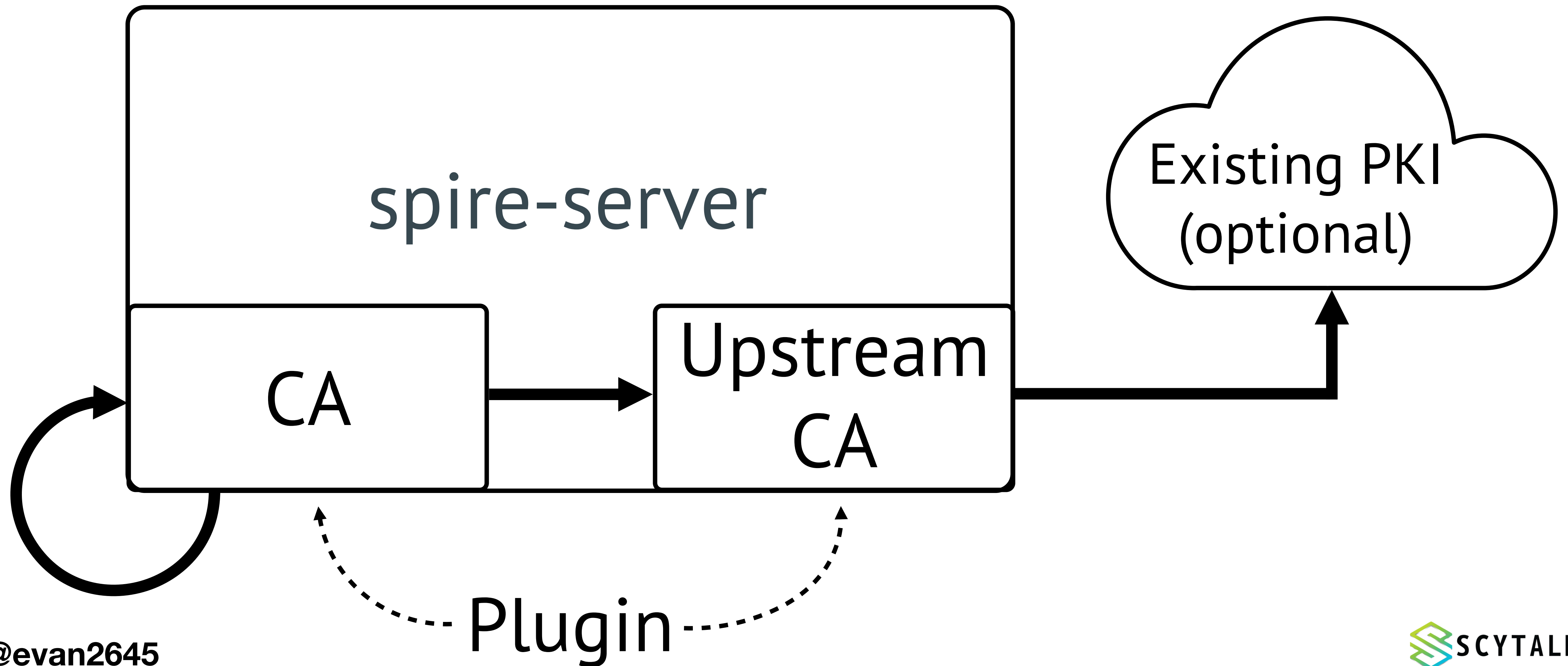
spire-agent

- Workload Attestation
- Workload API

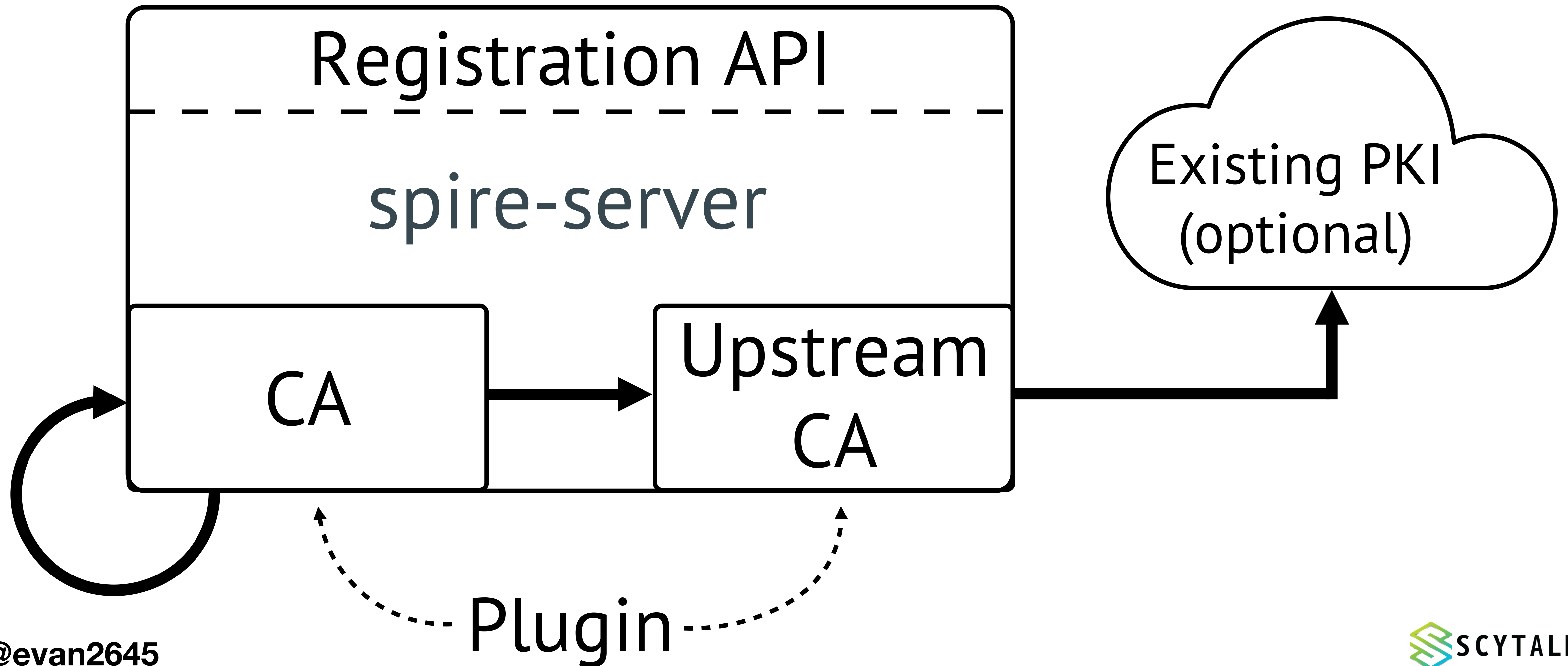
SPIRE Walkthrough



SPIRE Walkthrough



SPIRE Walkthrough



SPIRE Walkthrough

Example Identity Mapping

Parent ID: `spiffe://example.org/k8s/cluster/foo`

Selector: `k8s:ns:operations`

Selector: `k8s:sa:mediawiki`

Selector: `docker:image-id:746b819f315e`

SPIFFE ID: `spiffe://example.org/ops/wiki`

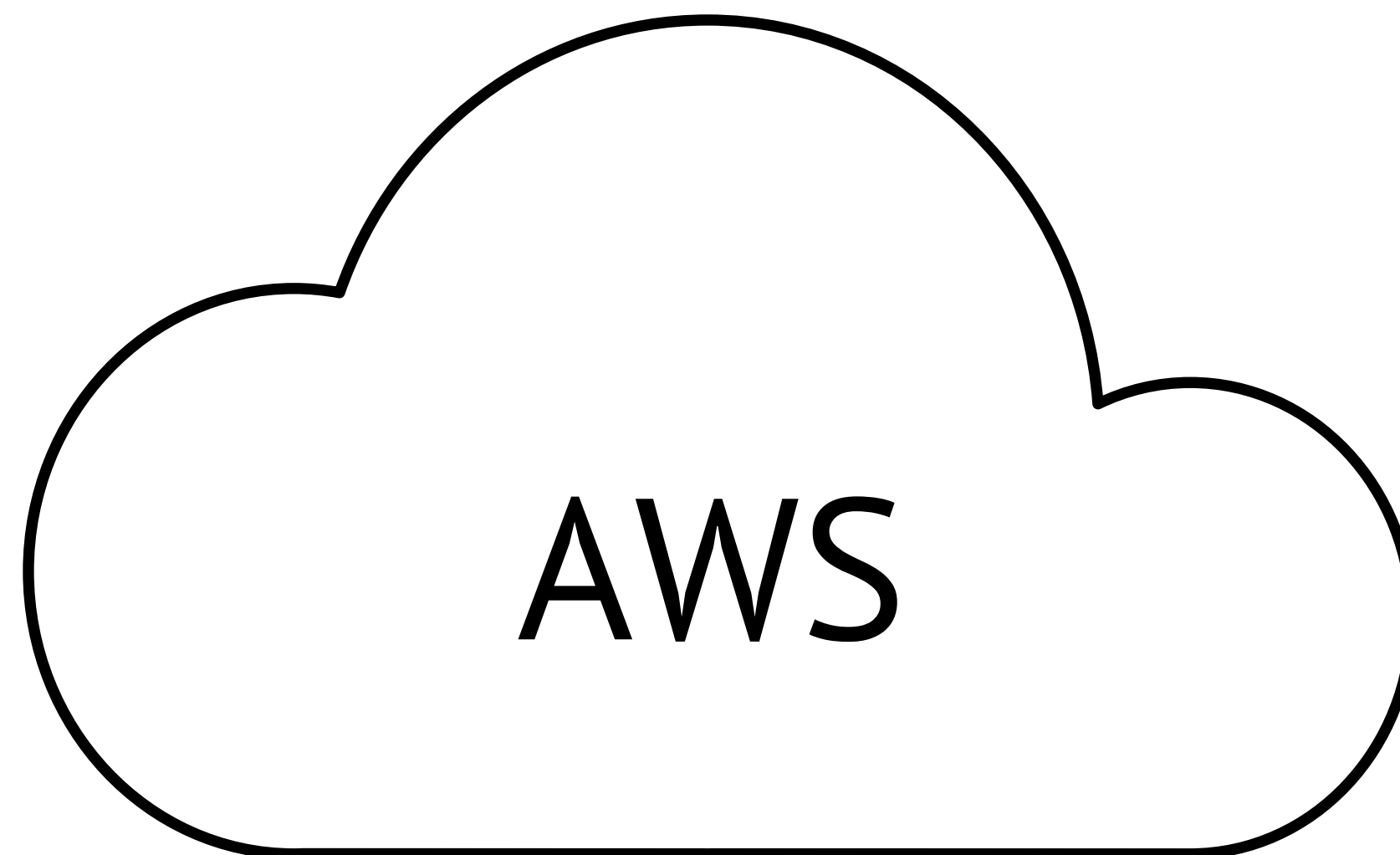
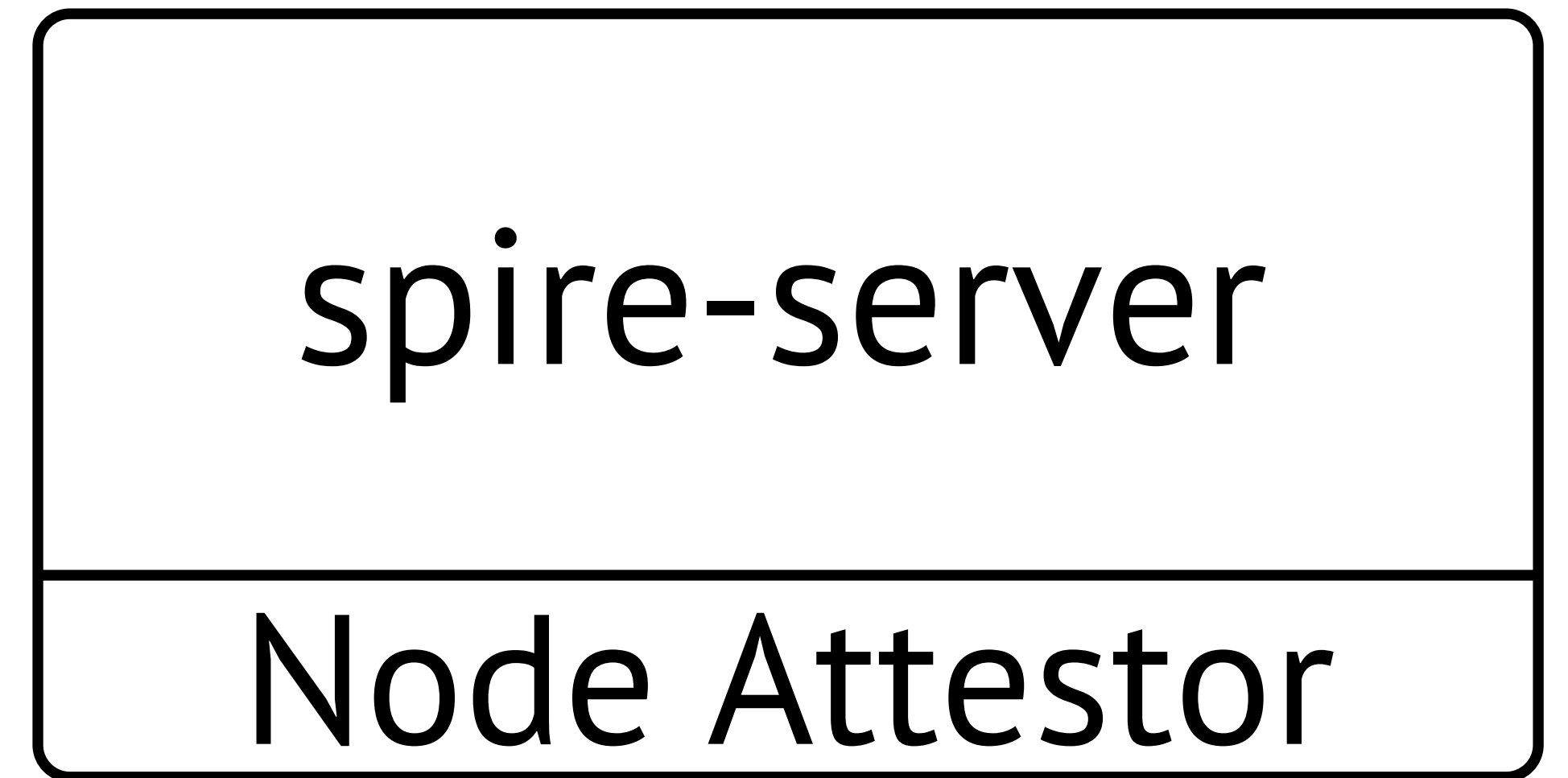
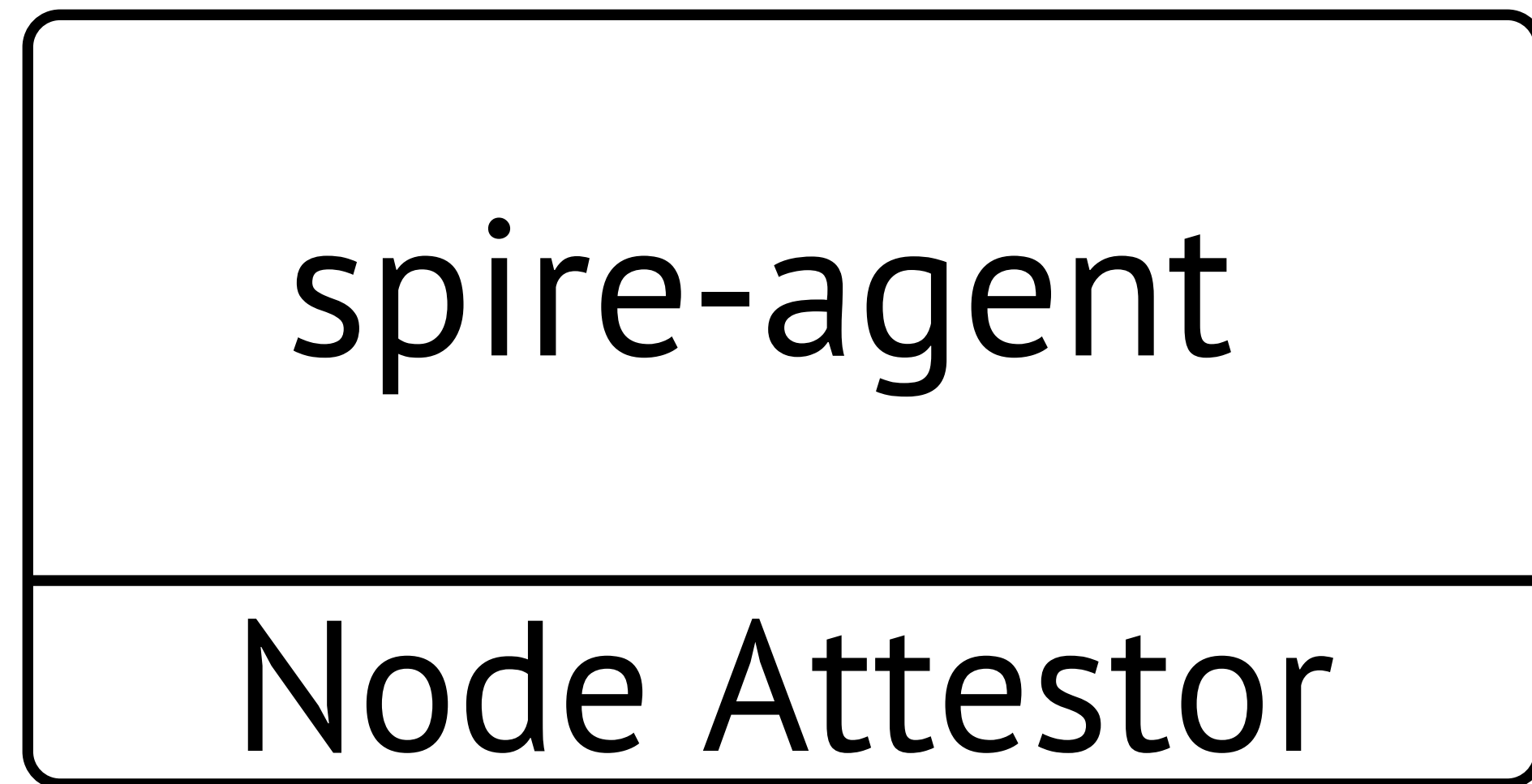
SPIRE Walkthrough

spire-server

Node Attestor

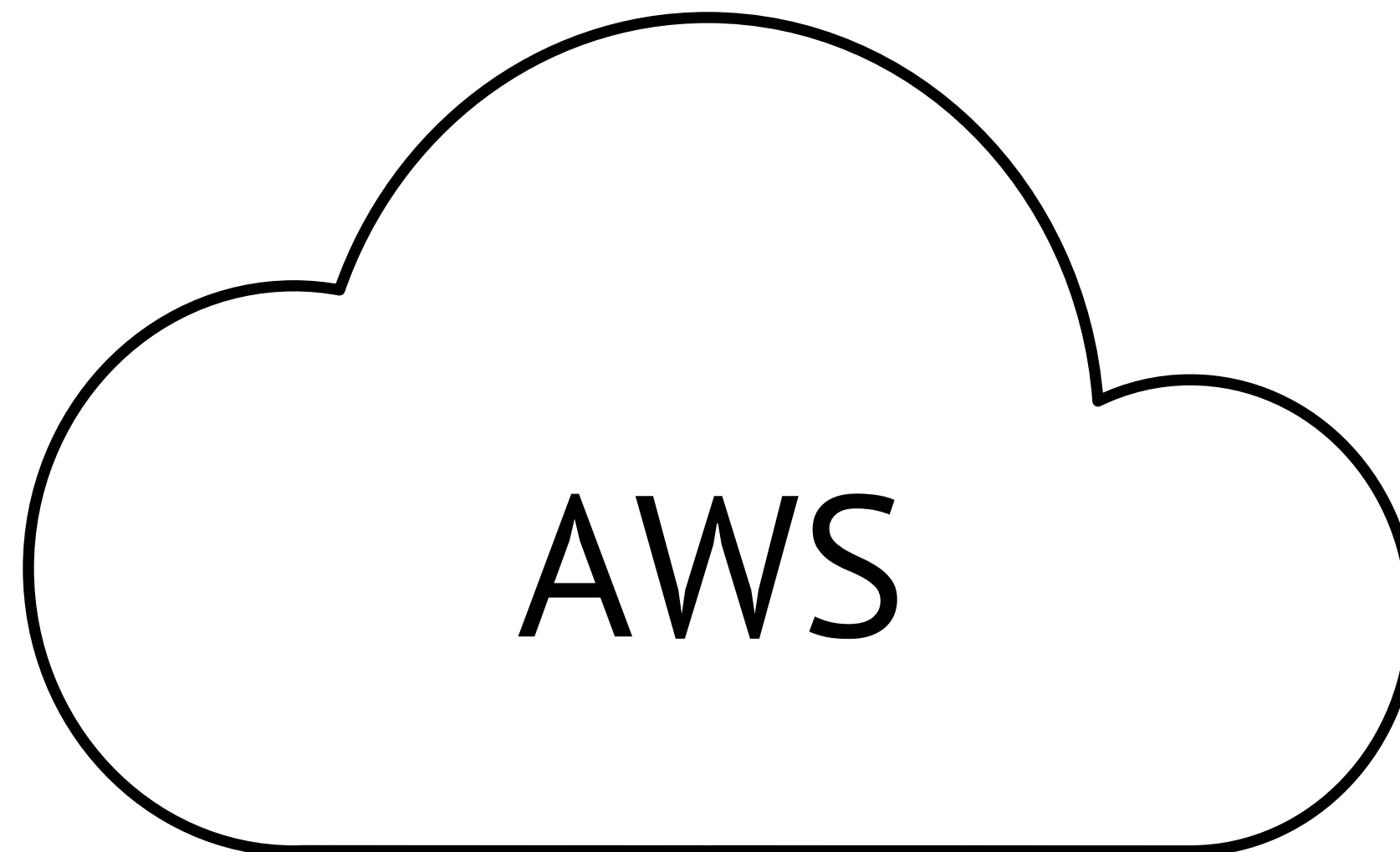
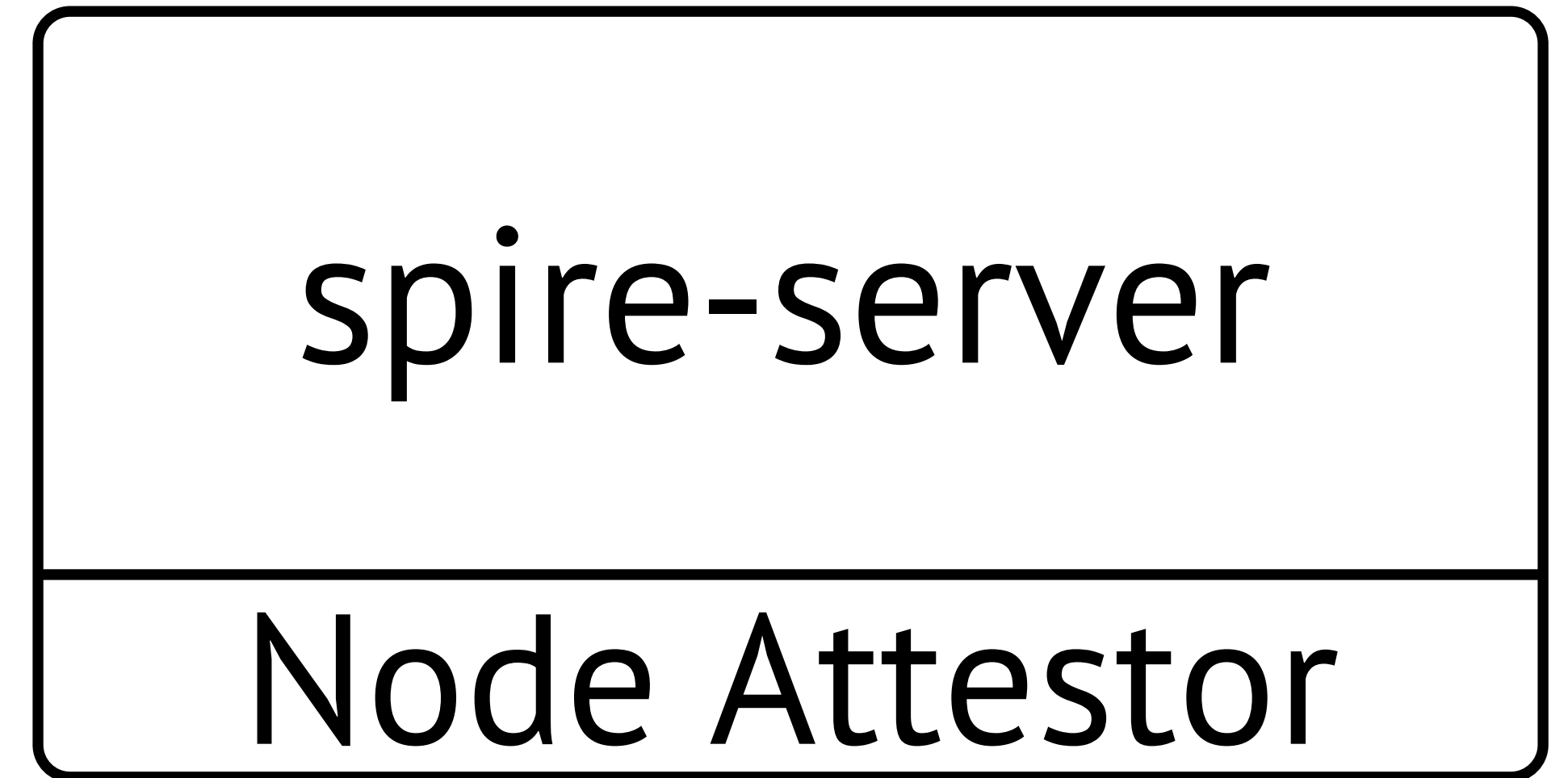
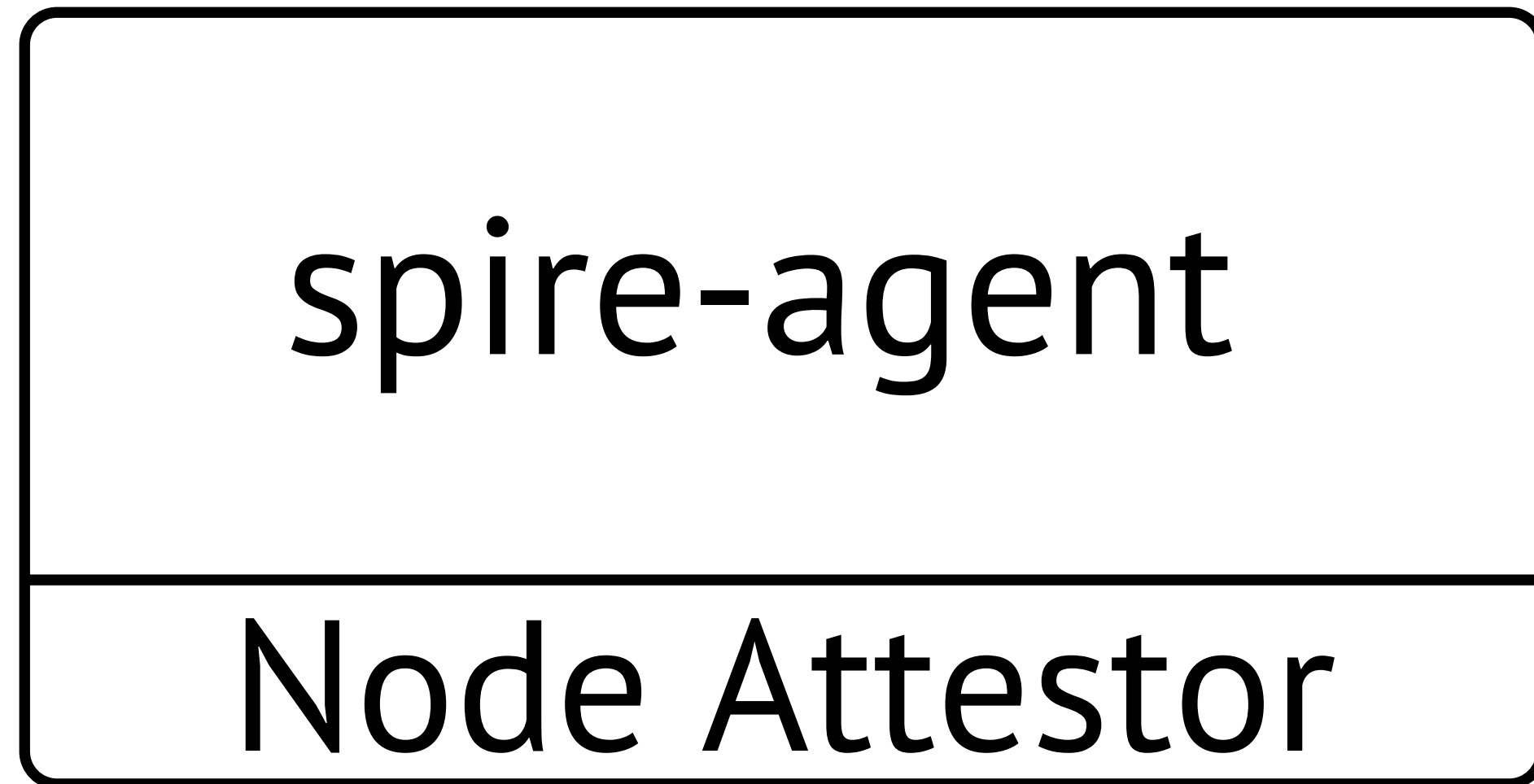
AWS

SPIRE Walkthrough



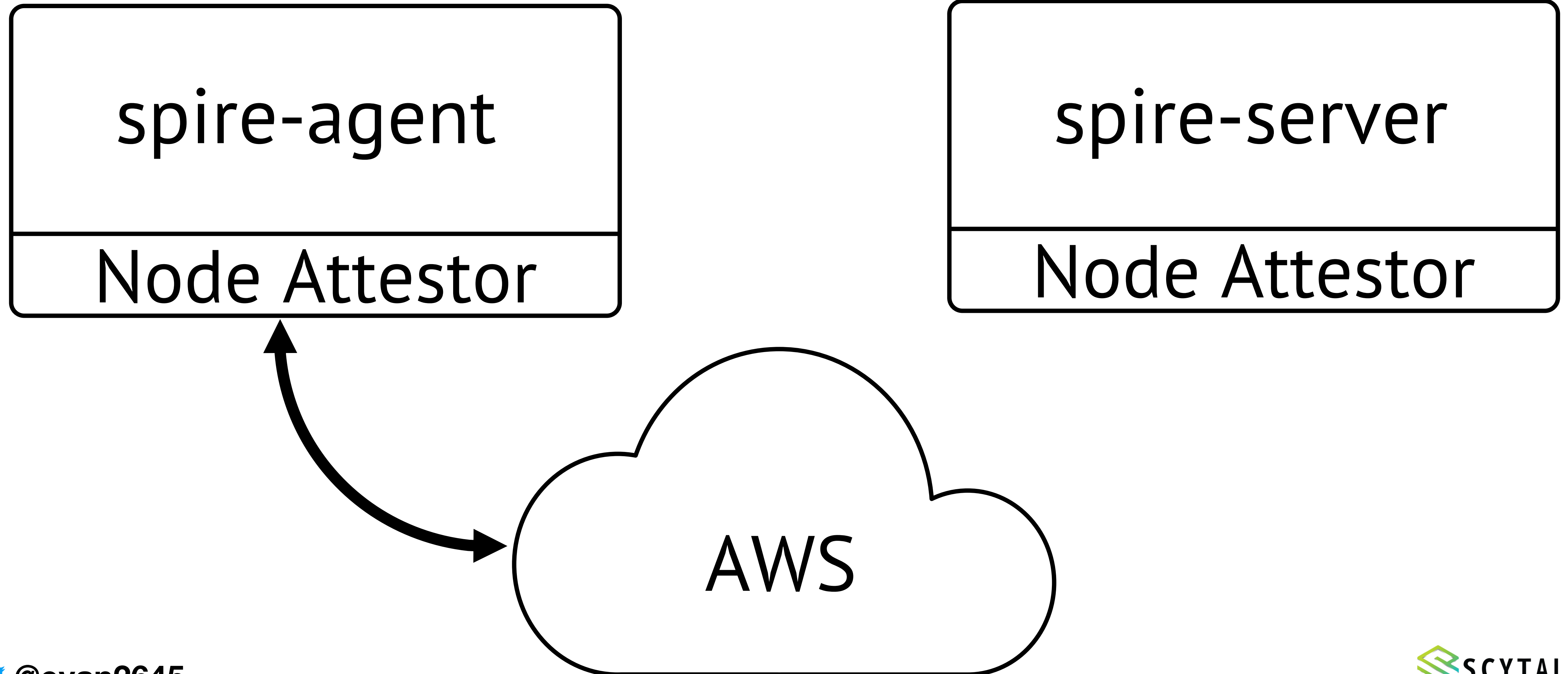
SPIRE Walkthrough

Agent Gathers
"Proof" of Identity



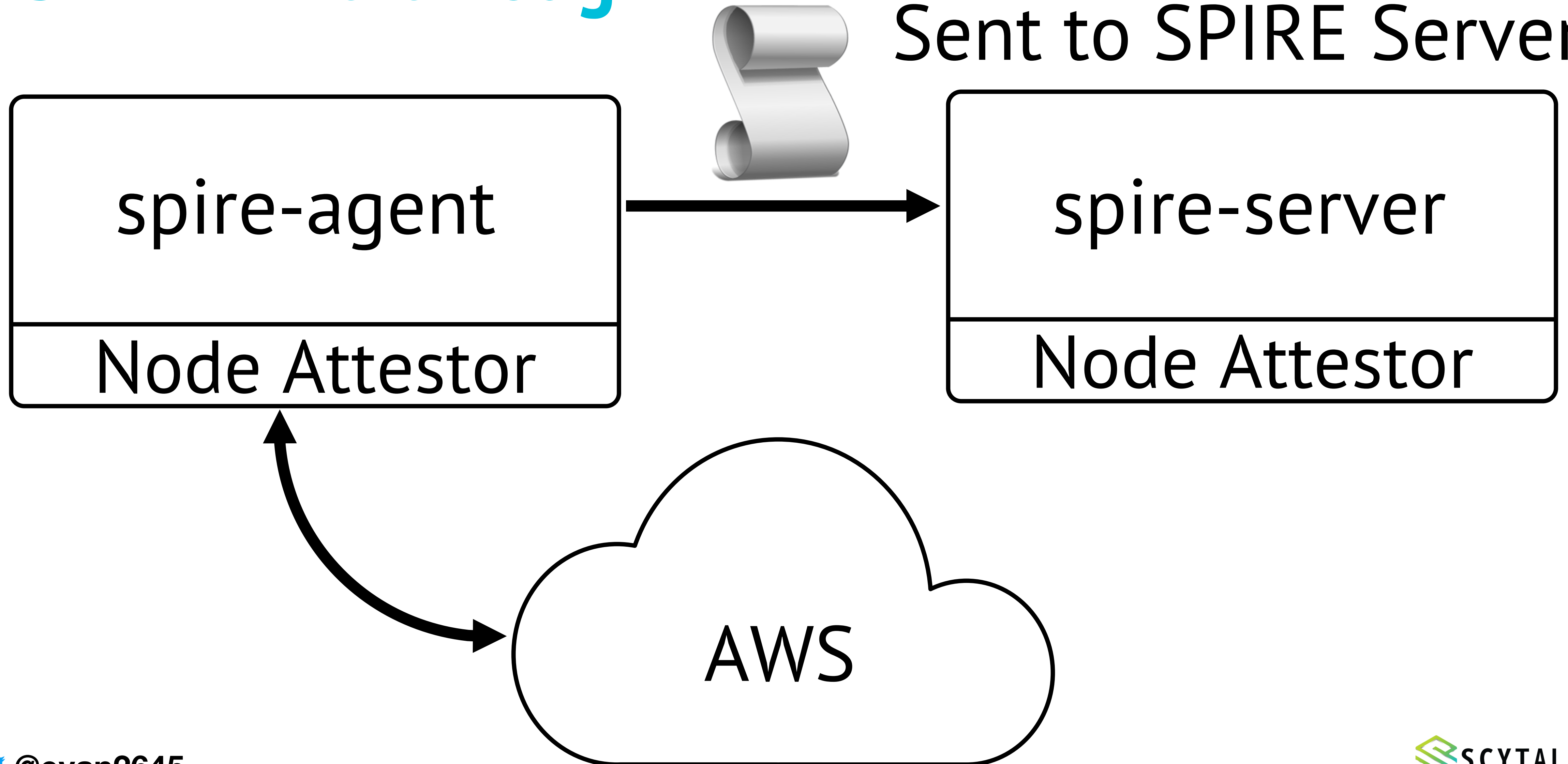
SPIRE Walkthrough

Agent Gathers "Proof" of Identity



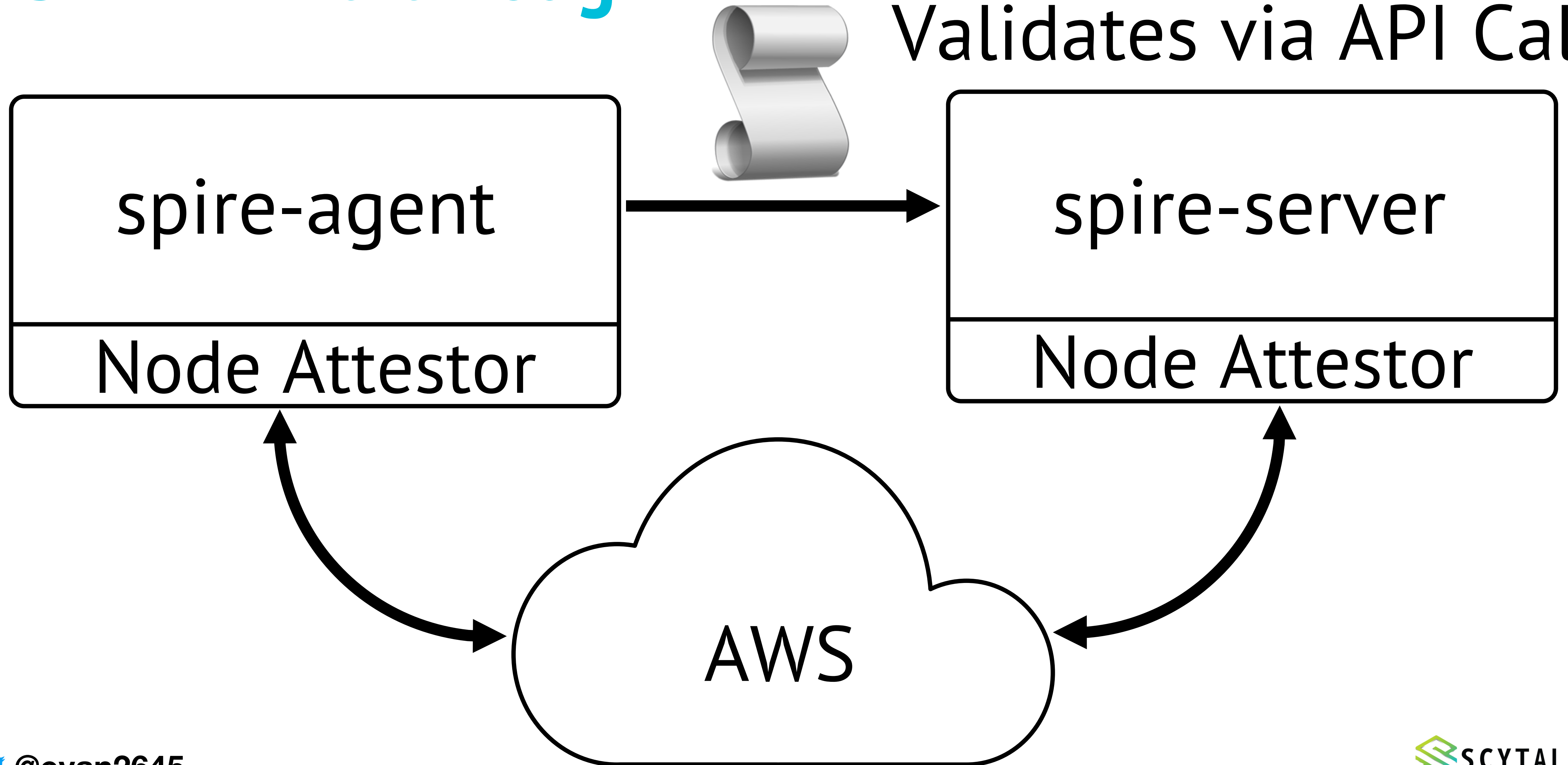
SPIRE Walkthrough

"Proof" of Identity
Sent to SPIRE Server



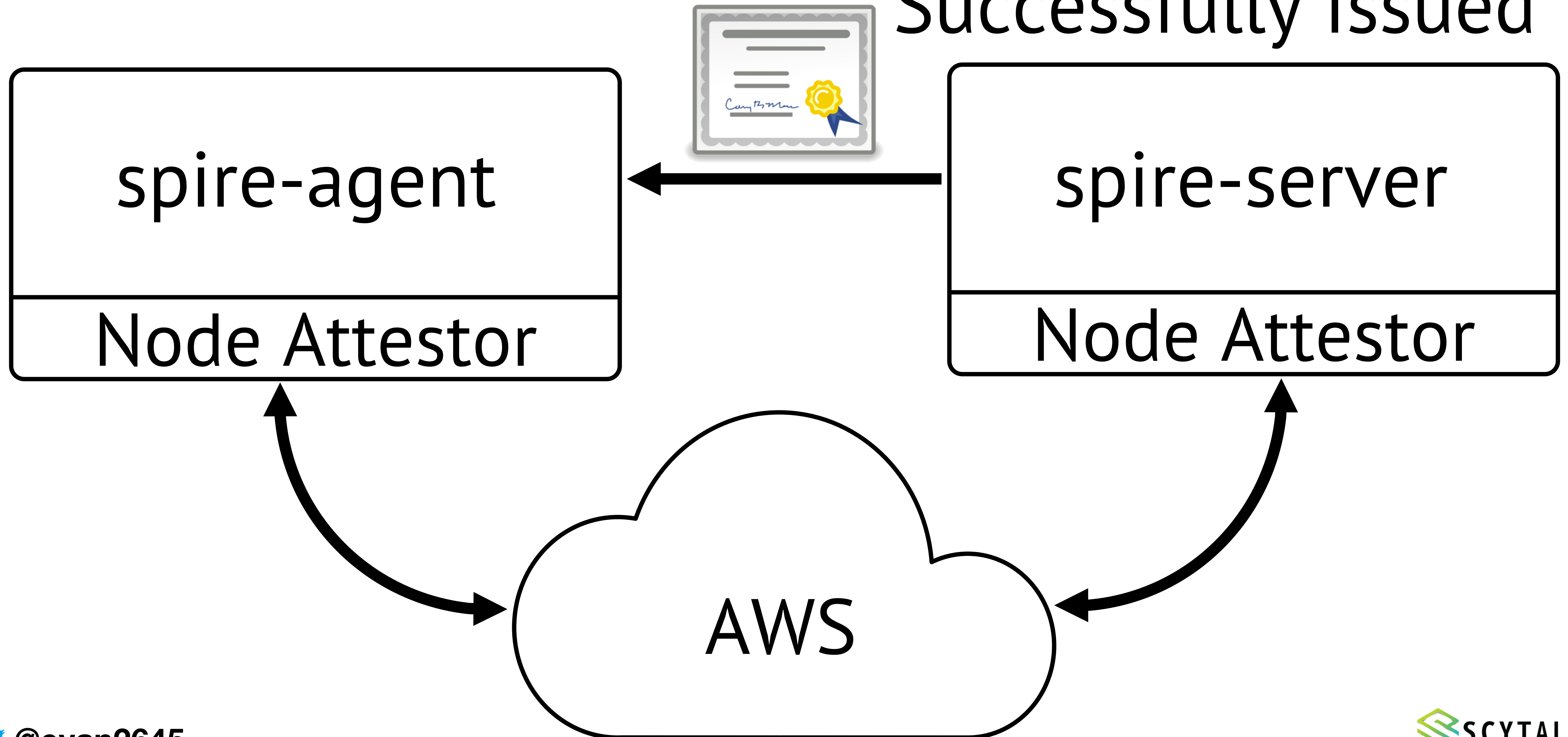
SPIRE Walkthrough

SPIRE Server
Validates via API Call



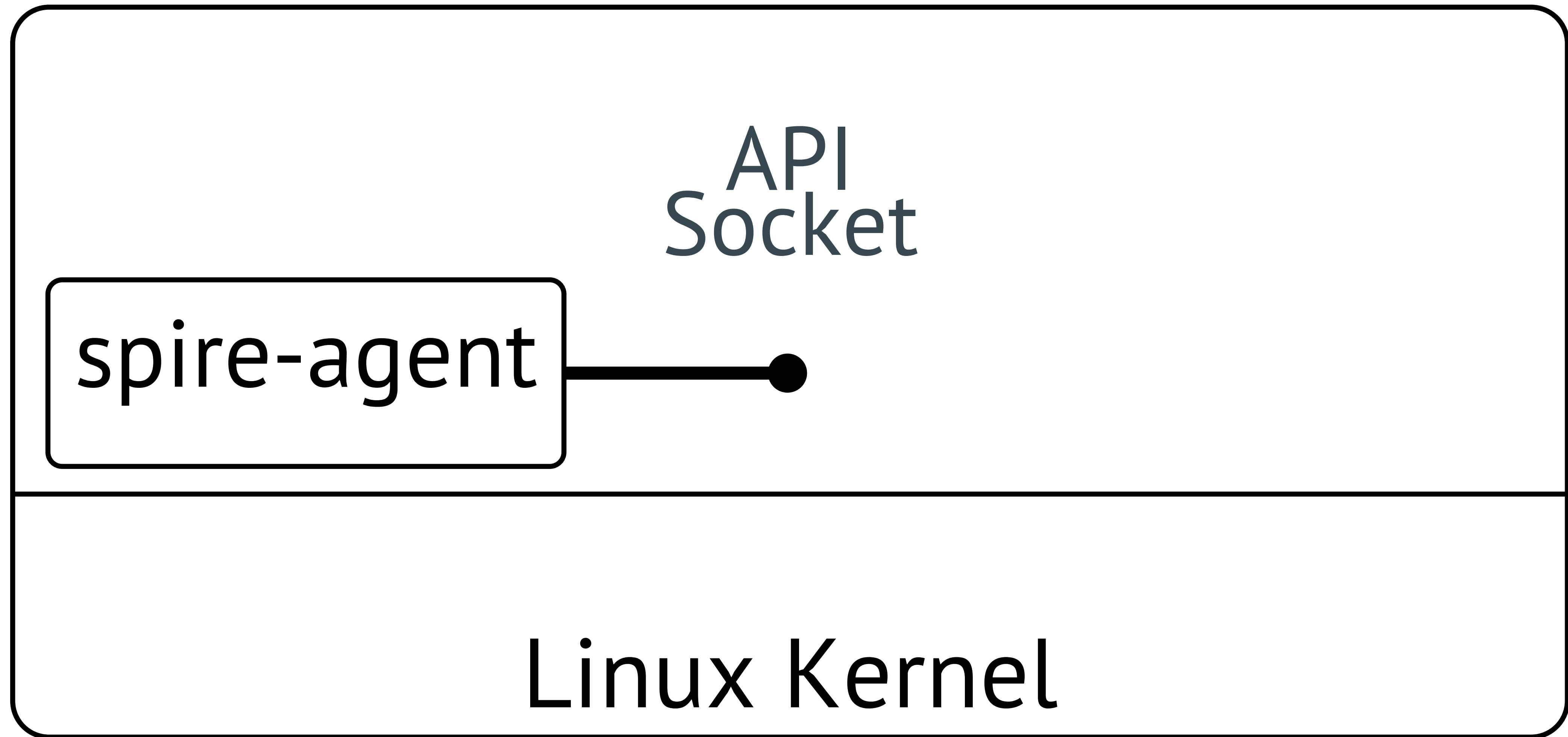
SPIRE Walkthrough

SPIFFE Identity is
Successfully Issued



SPIRE Walkthrough

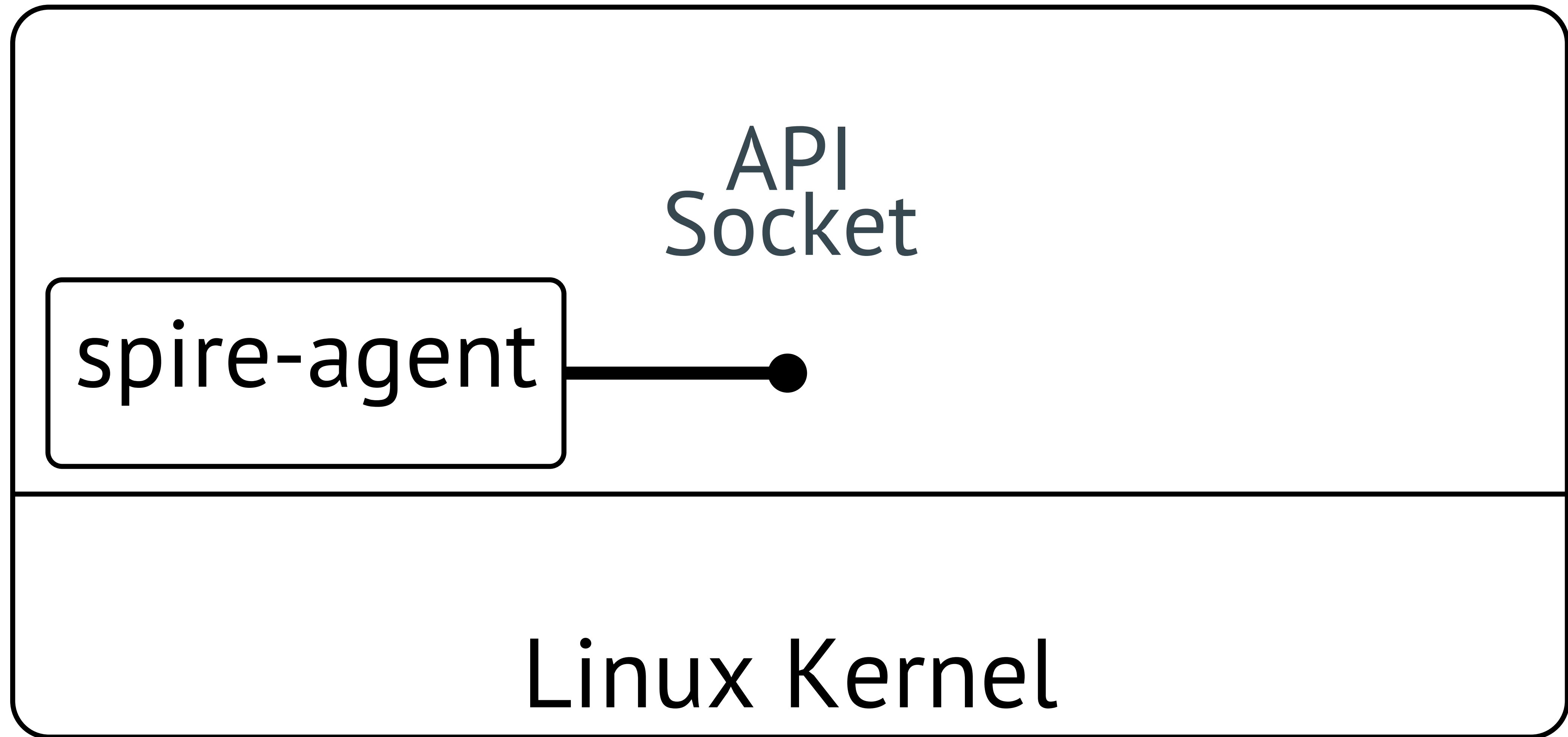
Server



Agent Exposes Workload API

SPIRE Walkthrough

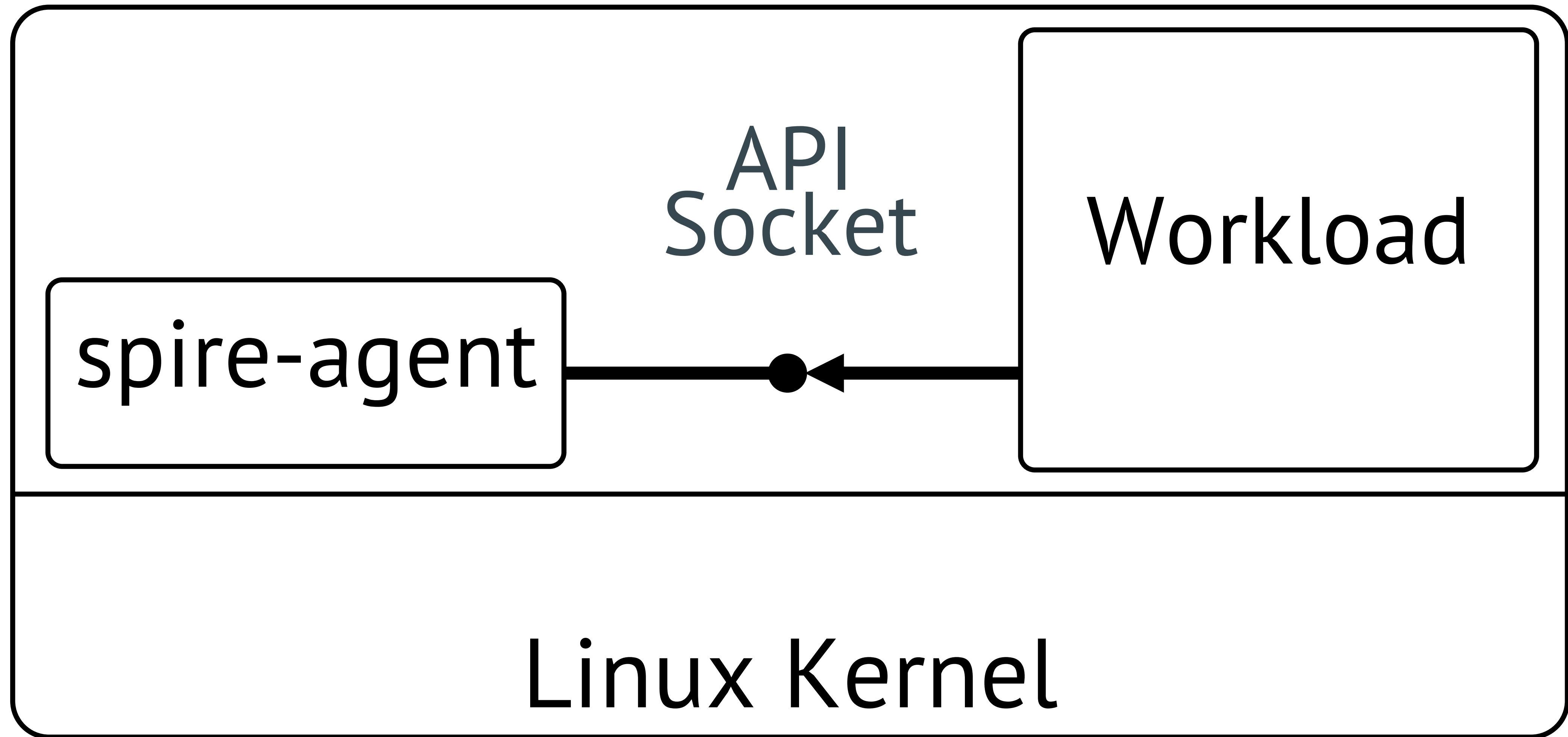
Server



Workload API is Unauthenticated

SPIRE Walkthrough

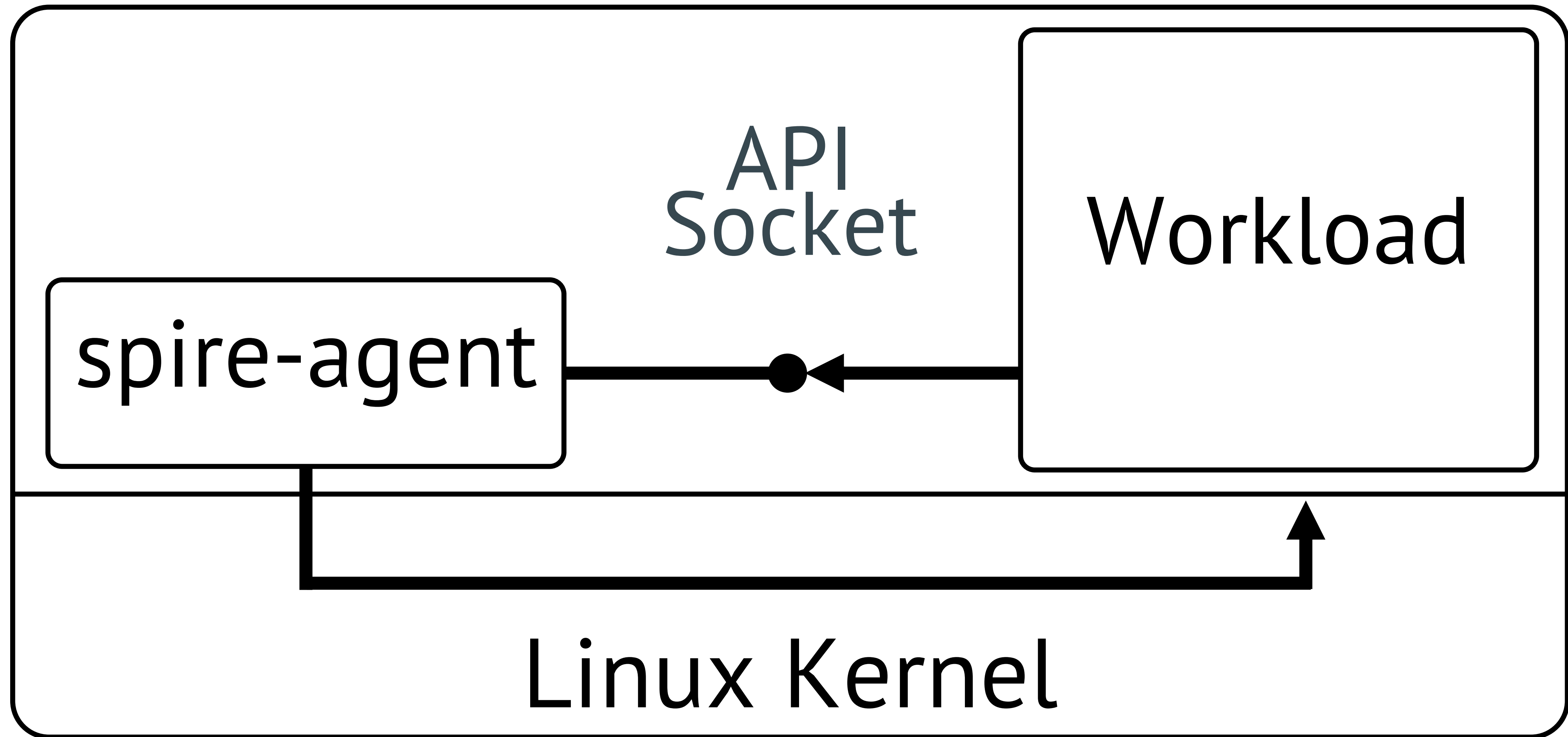
Server



Out-Of-Band Interrogation

SPIRE Walkthrough

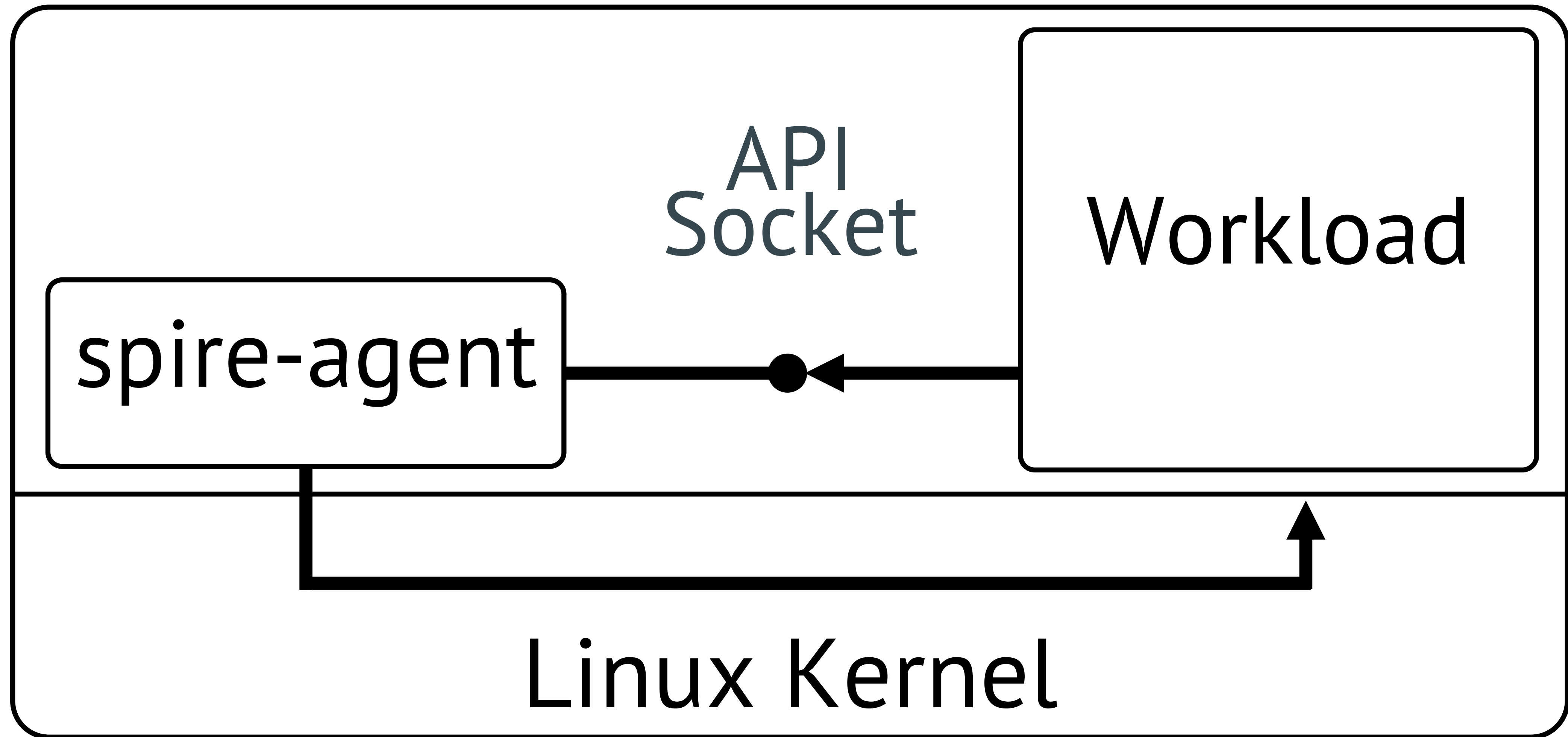
Server



Identify Process via Linux Syscall

SPIRE Walkthrough

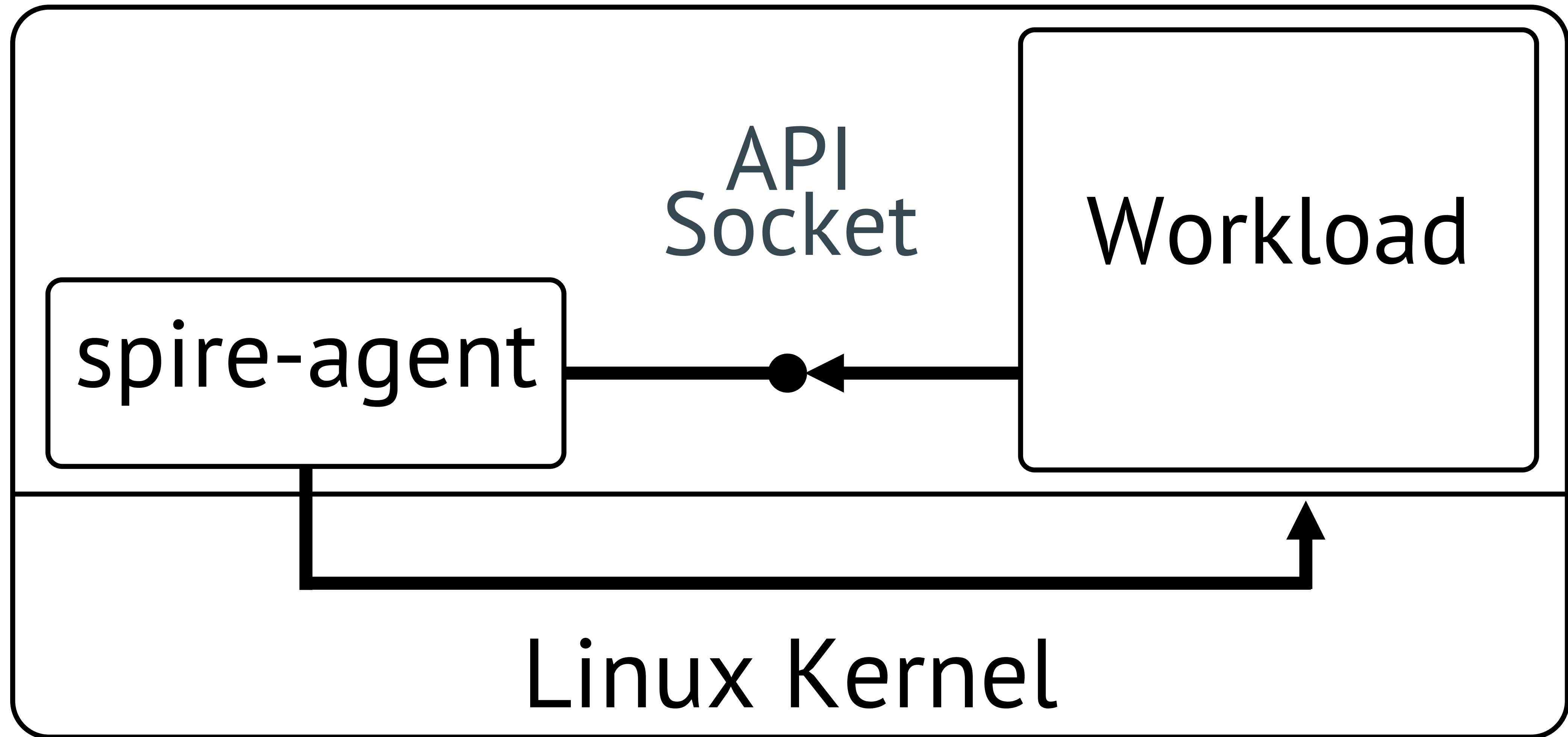
Server



Interrogation Logic is Pluggable

SPIRE Walkthrough

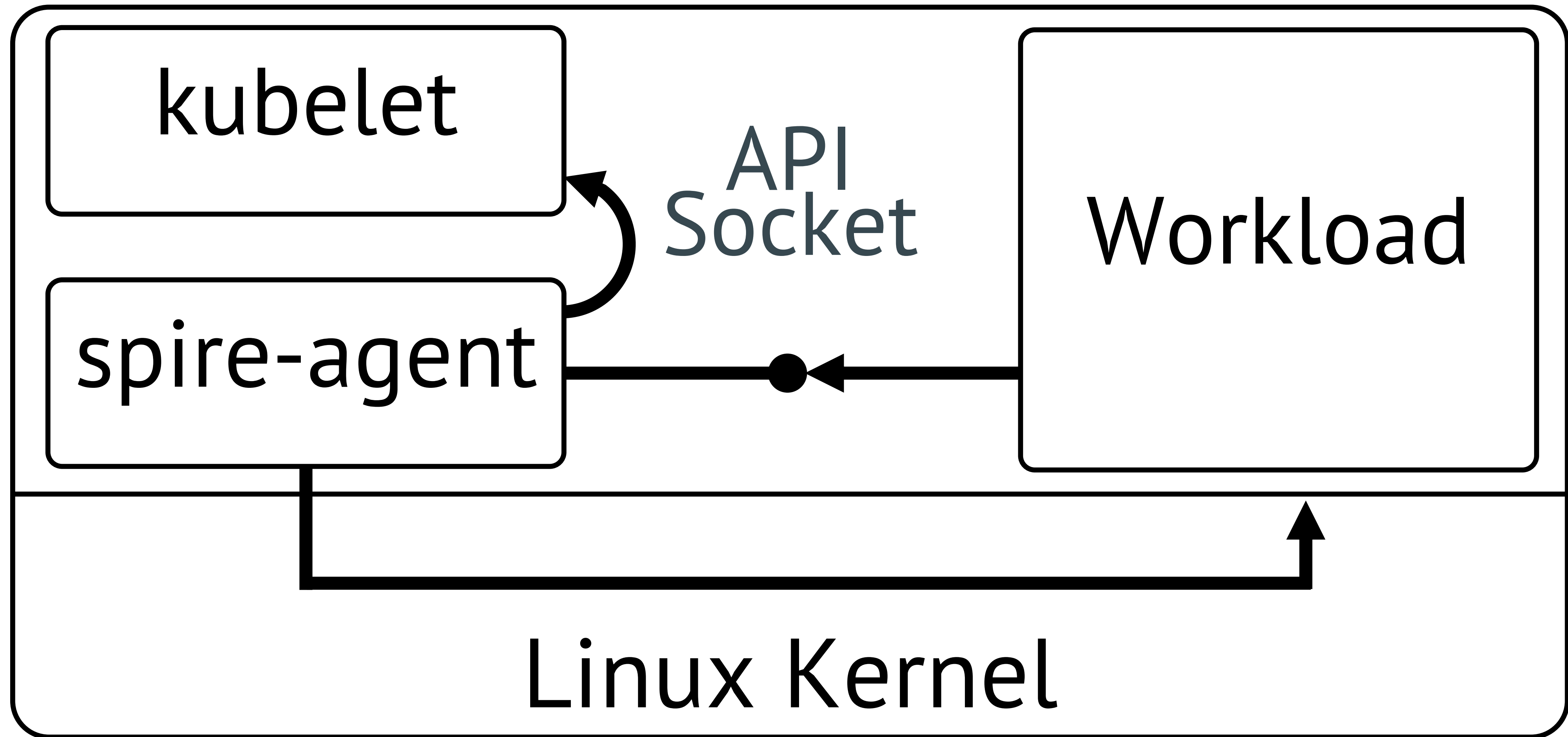
Server



Interrogation Logic is Pluggable

SPIRE Walkthrough

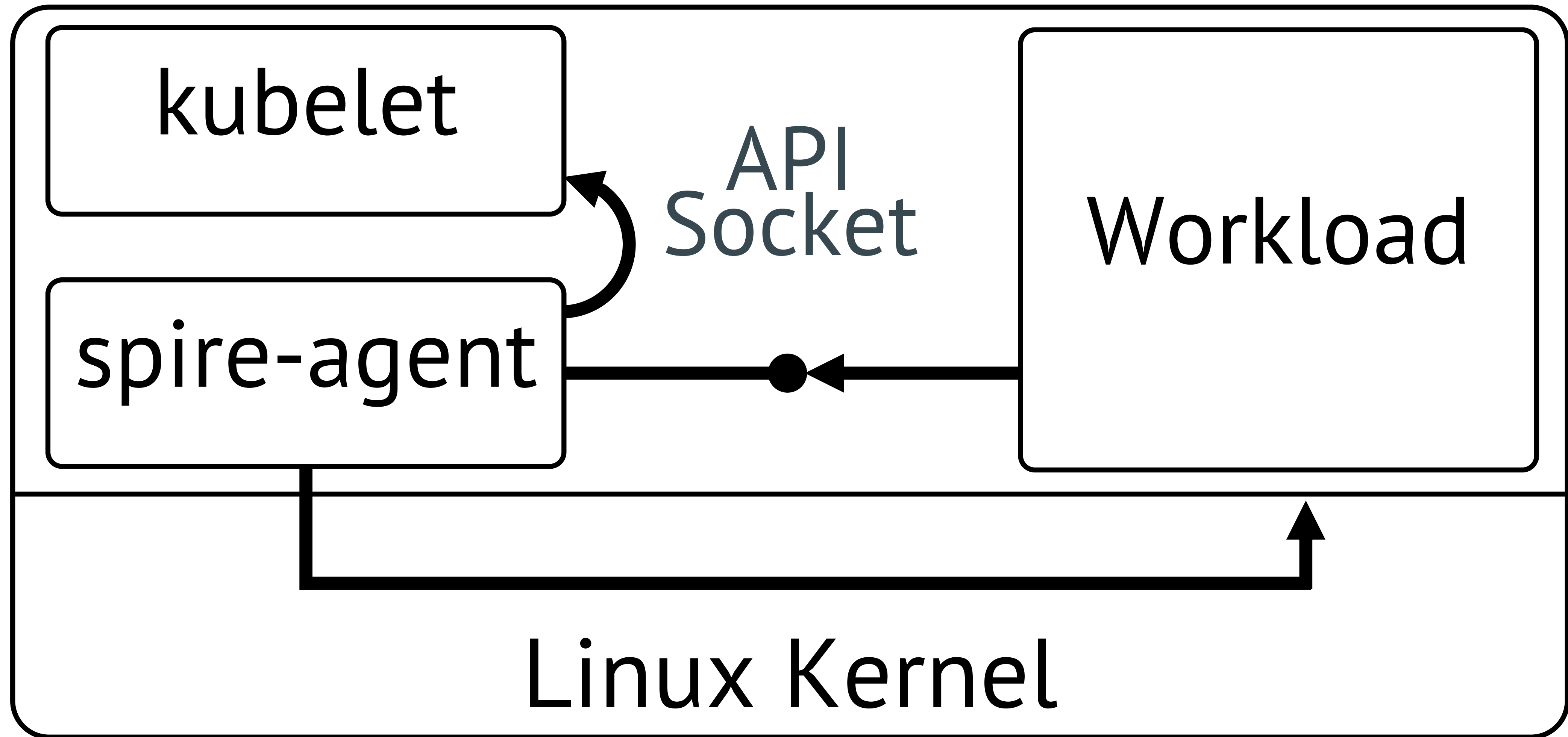
Server



Interrogation Logic is Pluggable

SPIRE Walkthrough

Server



@evan2645

AWS + Kernel: Trusted 3rd Party



SPIRE

Secure Introduction

SPIRE

Secure Introduction

In Summary

Looking Forward

Join Our Community

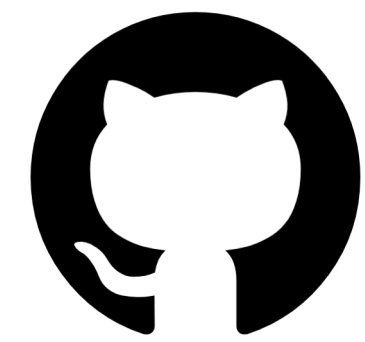
VERY Active Community

150+ Participants, 40+ Companies

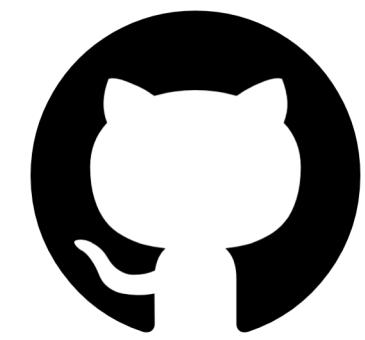
Special Interest Group Meetings

Active Standards + Software Development

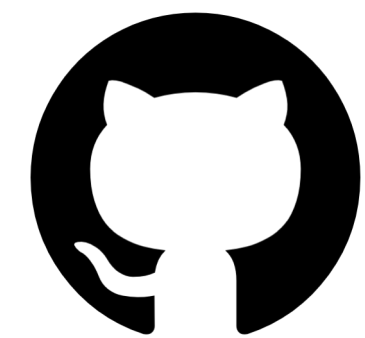
Play Today



[spiffe/spiffe](#)



[spiffe/spire](#)



[spiffe/spiffe-example](#)



[slack.spiffe.io](#)

Workload Identity

Evan Gilman

